

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_47057438-9c9\agent-a1bea65cdab0c3b6e.jsonl`  
- SHA-256: `b1b08e4f40c09285eca7193c73b03cb924573d44890bcfaece51eb622fc31fcf`  
- Source modified: `2026-06-16T03:05:55+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_47057438-9c9`  
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`
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Transcript

1. user (2026-06-16T03:01:27.114Z)

You are senior IP/privacy counsel writing an internal memo (general information, NOT a substitute for retained counsel) for an owner in Bangalore, India, who wants to COMMIT derived numeric data (shuttle trajectories, rally time-intervals, audio/person/court feature columns) from a labeled video corpus into a GitHub repo so external contributors can run regression tests WITHOUT the videos. Synthesize the research findings and the skeptical verifications below into ONE coherent, cited memo. Be explicit about the OWNED-corpus vs THIRD-PARTY-sourced fork, split the analysis by jurisdiction (India primary; US and EU because the product/repo are global), and state concrete, testable CONDITIONS under which a public commit of derived data is defensible (and which streams must NOT be published ? e.g. anything person/pose/biometric, anything involving minors). List the questions that genuinely need a retained lawyer. Use only citations that appear in the findings/verifications; do not invent any.

FINDINGS (JSON):

```
[  
  {  
    "question": "Is algorithmically-derived structured data ? per-frame shuttle (x,y) coordinates, rally start/end timestamps, audio-onset feature values ? extracted from a copyrighted sports video itself protected by copyright (uncopyrightable facts/data, or a \"derivative work\")? Does the derived numeric data inherit the video's copyright?",  
    "scope": "US copyright law analysis covering: idea/expression dichotomy (17 USC 102(b)); Feist (facts not copyrightable; sweat-of-the-brow rejected; originality in selection/arrangement); the 17 USC 101 \"derivative work\" definition; and intermediate/non-expressive copying + fair use (Google v. Oracle, Authors Guild v. Google, Sega v. Accolade, plus 2025 AI-training cases). This is legal-research synthesis of statutes, case law, and the U.S. Copyright Office Fair Use Index ? NOT formal legal advice. Focus is the copyright status of the OUTPUT data itself; the separate question of whether the EXTRACTION ACT (transiently copying the video) infringes is addressed via the intermediate-copying line. Foreign sui generis database rights (e.g., the EU Database Directive) are out of scope but flagged.",  
    "findings": [  
      {  
        "claim": "Facts are categorically not copyrightable because they are discovered, not authored, and thus lack the originality the Constitution and Copyright Act require; effort ('sweat of the brow') does not earn copyright. This is the foundational holding most directly governing extracted numeric data.",  
        "support": "Feist Publications v. Rural Telephone Service Co., 499 U.S. 340 (1991): the Court held 'facts are not copyrightable' because they 'do not owe their origin to an act of authorship' and so are not original, and expressly held that 'originality, not sweat of the brow, is the touchstone of copyright protection.' Copyright 'rewards originality, not effort.' Per-frame (x,y) shuttle positions, rally start/end timestamps, and audio-onset values are observable physical facts about events recorded in the video ? discovered measurements, not authored expression ? so they fall squarely within the unprotected 'facts' category regardless of the computational effort to derive them.",  
        "sourceName": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? Justia U.S. Supreme Court Center",
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```

    "sourceUrl": "https://supreme.justia.com/cases/federal/us/499/340/",
    "confidence": "high"
  },
  {
    "claim": "Facts and statistics GENERATED BY A SPORTING EVENT are not copyrightable even though the audiovisual BROADCAST of that event is ? the closest factual analogue to extracting shuttle coordinates and rally timings from a match video.",
    "support": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir. 1997): the court held 'Sports events are not authored in any common sense of the word' and that the underlying scores/statistics are uncopyrightable facts ('purely factual information which any patron of an NBA game could acquire from the arena'), while only the broadcast ? the creative work of directors, camera operators, editors ? is protected. By direct analogy, shuttle positions, rally boundaries, and onset values are facts ABOUT the game that any spectator could in principle observe; they are not original to whoever created the video, so they do not inherit the video's copyright.",
    "sourceName": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir. 1997) ? Wikipedia case summary",
    "sourceUrl": "https://en.wikipedia.org/wiki/National_Basketball_Ass%27n_v._Motorola,_Inc.",
    "confidence": "high"
  },
  {
    "claim": "Copyright never extends to any idea, procedure, process, system, method of operation, concept, principle, or discovery ? only to original expression. Measured coordinates and timestamps are 'discoveries'/facts about the recorded event and fall in the §102(b) exclusion.",
    "support": "17 U.S.C. § 102(b): 'In no case does copyright protection for an original work of authorship extend to any idea, procedure, process, system, method of operation, concept, principle, or discovery, regardless of the form in which it is described, explained, illustrated, or embodied in such work.' This is the statutory codification of the idea/expression dichotomy and confirms that the factual/discovered content extracted from the video (the where-and-when of the shuttle and rallies) is outside the scope of the video's copyright.",
    "sourceName": "17 U.S.C. § 102 ? BitLaw / U.S. Code",
    "sourceUrl": "https://www.bitlaw.com/source/17usc/102.html",
    "confidence": "high"
  },
  {
    "claim": "A 'derivative work' is a work that recasts, transforms, or adapts a preexisting work's EXPRESSION (e.g., translation, motion-picture version, abridgment). Stripping out non-expressive numeric facts is not 'recasting' the audiovisual expression, so the coordinate/timestamp/feature data is not a derivative work that inherits the video's copyright.",
    "support": "17 U.S.C. § 101 defines a derivative work as 'a work based upon one or more preexisting works, such as a translation, musical arrangement, dramatization, fictionalization, motion picture version, sound recording, art reproduction, abridgment, condensation, or any other form in which a work may be recast, transformed, or adapted.' Each enumerated example reuses the original's protected EXPRESSION. A table of (x,y) positions and timestamps reproduces none of the video's expressive elements (framing, editing, commentary, imagery); it captures only facts, which §102(b)/Feist place outside copyright. A derivative-work claim therefore fails because there is no protected expression carried into the output.",
    "sourceName": "17 U.S.C. § 101 Definitions ? Legal Information Institute (Cornell)",
    "sourceUrl": "https://www.law.cornell.edu/uscode/text/17/101",
    "confidence": "high"
  },
  {
    "claim": "A compilation/database of facts can earn a THIN copyright, but only in the original SELECTION, COORDINATION, OR ARRANGEMENT ? never in the underlying facts. So a creatively-organized dataset might get limited protection for its structure, but a competitor remains free to copy the raw numbers themselves.",
    "support": "17 U.S.C. § 101 defines a 'compilation' as data 'selected, coordinated, or arranged in such a way that the resulting work as a whole constitutes an original work of authorship,' and Feist holds copyright in a compilation 'extends only to those components of the work that are original to the author, not to the facts themselves.' Practical upshot: any copyright the extractor obtains is in the dataset's expressive arrangement (if creative), not in

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the shuttle coordinates, rally timestamps, or onset values, which remain free for others to use.",

"sourceName": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? Justia",

"sourceUrl": "https://supreme.justia.com/cases/federal/us/499/340/",

"confidence": "high"

},

{

"claim": "The act of EXTRACTION (transiently copying the video to analyze it) is a distinct question from the status of the OUTPUT data ? and intermediate copying CAN infringe even when the end product does not reproduce protected expression, so the extraction step must independently rely on a license or fair use.",

"support": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992): 'intermediate copying of computer object code may infringe the exclusive rights granted to the copyright owner in section 106 ... regardless of whether the end product of the copying also infringes those rights.' But the same court held that copying to access UNPROTECTED functional/ideational elements is fair use where it is the only means of access and there is a legitimate reason, because §102(b) elements 'are not protected by copyright.' Applied here: even though the numeric output is unprotectable, ingesting the video to compute it is itself a reproduction that needs a lawful basis (license to the source video, or a fair-use justification for the analytic copying).",

"sourceName": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992) ? BitLaw",

"sourceUrl": "https://www.bitlaw.com/source/cases/copyright/Sega-Accolade.html",

"confidence": "high"

},

{

"claim": "Copying a whole work for a NON-EXPRESSIVE, analytical purpose ? to build an index/search function or extract data rather than to deliver the original's expression to the public ? is strongly favored as transformative fair use. This squarely supports the extraction pipeline if it must rely on fair use.",

"support": "Authors Guild v. Google, Inc., 804 F.3d 202 (2d Cir. 2015): the court held that 'verbatim copying of works for a different, non-expressive purpose can be a transformative fair use' ? Google copied entire books to build a searchable index and show snippets, which 'augments public knowledge ... without providing the public with a substantial substitute' for the originals. Extracting positional/temporal/audio-feature data is non-expressive analytic use that does not republish the video's expression, fitting this transformative-purpose rationale.",

"sourceName": "Authors Guild v. Google, Inc., 804 F.3d 202 (2d Cir. 2015) ? Justia (Second Circuit opinion)",

"sourceUrl": "https://law.justia.com/cases/federal/appellate-courts/ca2/13-4829/13-4829-2015-10-16.html",

"confidence": "high"

},

{

"claim": "The Supreme Court has reinforced that functional, non-expressive material lies far from copyright's core and that reusing it for a new, productive purpose is transformative fair use ? and that any losses tied to function rather than expression do not count against fair use.",

"support": "Google LLC v. Oracle Am., Inc., 141 S. Ct. 1183 (2021), as summarized in the U.S. Copyright Office Fair Use Index: the Court found Google's copying of declaring code 'transformative' because it 'adds something new and important,' that the code was 'functional in nature' and 'further than most computer programs from the core of copyright,' and that Oracle's losses 'were due not to the expressive qualities ... but to programmers having become accustomed to' certain calls. By analogy, the value of extracted shuttle/rally/audio data derives from factual/functional content, not the video's expression, reinforcing that the data does not inherit the video's copyright and that analytic extraction is transformative.",

"sourceName": "Google LLC v. Oracle Am., Inc., 141 S. Ct. 1183 (2021) ? U.S. Copyright Office Fair Use Index summary",

"sourceUrl": "https://www.copyright.gov/fair-use/summaries/google-llc-oracle-am-inc-2021.pdf",

"confidence": "high"

},

{

"claim": "Recent (2025) AI-training decisions confirm the modern through-line: courts treat using copyrighted works 'as data rather than expression' as transformative/non-expressive

use that favors fair use, while output that reproduces a work's expressive market function does not.",

"support": "As summarized in legal commentary on Bartz v. Anthropic, Kadrey v. Meta, and Thomson Reuters v. Ross (2025): courts 'accept complete copying when it is technologically necessary and non-expressive, with the end user never receiving the protected content,' and 'when the training is analytical ? using works as data rather than expression ? it passes' fair use. This supports treating feature extraction (data, not expression) as non-infringing in its OUTPUT and favorable on fair use for the extraction step; however, this body of law is new, fact-specific, and partly unsettled (mixed outcomes; appeals likely).",

"sourceName": "Training Data on Trial: AI's First Fair Use Test ? IPWatchdog",

"sourceUrl": "https://ipwatchdog.com/2025/10/09/training-data-trial-ai-first-fair-use-test/",

"confidence": "medium"

}

],

"bottomLine": "Under US copyright law, the DERIVED NUMERIC DATA itself does NOT inherit the video's copyright. Per-frame shuttle (x,y) coordinates, rally start/end timestamps, and audio-onset feature values are uncopyrightable FACTS/measurements about the recorded event ? discovered, not authored ? so under Feist (facts are never copyrightable; \"sweat of the brow\" rejected), §102(b) (no copyright in any \"idea, procedure, process ... or discovery\"), and NBA v. Motorola (facts generated by a sporting event are uncopyrightable even though the broadcast is), the data is outside the scope of the video's copyright. It is also NOT a \"derivative work\" under §101: a derivative work recasts/transforms/adapts the original's protected EXPRESSION (framing, editing, imagery, commentary), and a coordinate/timestamp table carries none of that expression ? only facts. The most that copyright could attach to is a THIN protection in an original SELECTION/COORDINATION/ARRANGEMENT of the resulting dataset (a creatively designed schema), never in the raw numbers, which competitors stay free to copy. CRITICAL CAVEAT ? keep two questions separate: (1) the STATUS of the output data (clearly unprotected facts), versus (2) the ACT of extraction. Sega v. Accolade holds intermediate copying CAN infringe even when the end product reproduces no protected expression, so transiently copying/ingesting the source video to compute the data needs its own lawful basis ? either a license to that video or a fair-use defense. The fair-use line is strongly favorable for analytic, non-expressive extraction: Authors Guild v. Google (verbatim copying for a non-expressive search/index purpose is transformative fair use), Google v. Oracle (functional material is far from copyright's core; reuse for a new productive purpose is transformative), and the 2025 AI-training rulings (using works \"as data rather than expression\" favors fair use). Net: ingest only videos you have the right to use (clean-licensed/owned/permissioned sources), and the derived numeric dataset is yours to use and publish as uncopyrightable facts ? consistent with the repo's \"paid LLMs are inference-only, never a training-data source\" and \"commercial-clean licensing\" guardrails, which address the SOURCE-acquisition side that this fact/expression analysis does not resolve.",

"caveats": [

"NOT legal advice ? this is a research synthesis of US statutes/case law and the U.S. Copyright Office Fair Use Index. For a commercial product, get a qualified IP attorney's opinion, especially on the extraction/ingestion step.",

"Two distinct questions: (a) the COPYRIGHT STATUS of the output data (settled: unprotectable facts) versus (b) whether the ACT of copying the source video to extract it infringes. This brief answers (a) firmly; (b) depends on your license to the source video or a fair-use defense, which is fact-specific.",

"Fair use for the extraction step is a multi-factor, case-by-case defense, not a categorical safe harbor. The 2025 AI-training decisions (Bartz v. Anthropic, Kadrey v. Meta, Thomson Reuters v. Ross) are recent, fact-bound, partly conflicting, and likely subject to appeal ? treat their support as directional, not settled (rated medium confidence).",

"A creative SELECTION/ARRANGEMENT of the dataset (schema, curation, an annotated golden set) could attract THIN compilation copyright in its structure ? but that protects only the arrangement, never the underlying numbers, and a purely mechanical/exhaustive layout (every frame, every rally) likely has too little originality to qualify (the Feist 'garden-variety alphabetical listing' problem).",

"This is US law. Other jurisdictions differ materially ? notably the EU, where the sui generis Database Right (Database Directive 96/9/EC) can protect a database built with substantial investment EVEN absent original selection/arrangement. If you operate or distribute in the EU/UK, raw extracted data may enjoy protection that US law denies. Not researched here.",

"Justia's Feist page returned HTTP 403 to the fetcher, so Feist quotes are corroborated via the search-result extract and the Justia URL rather than a full-text fetch; the Feist holdings cited are well established and also reflected in the §101/§102 statutory sources that

were fetched directly.",

"Adjacent doctrines not analyzed: trademark, contractual/Terms-of-Service restrictions on the source video, broadcaster anti-piracy and hot-news misappropriation (NBA v. Motorola also addressed a state misappropriation claim), and personality/biometric rights ? any of which could constrain use independent of copyright."

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    ],
    "webUsed": true
  },
  {
    "question": "Under the EU sui generis database right (Directive 96/9/EC) and copyright, does extracting and redistributing DERIVED match data (coordinates / event timestamps) from a broadcast infringe?",
    "scope": "EU law only (Directive 96/9/EC as interpreted by the CJEU). Covers: (1) sui generis database right ? \"obtaining vs creating\" data and \"extraction/re-utilisation of a substantial part\"; (2) database copyright (originality); (3) where re-utilisation occurs (targeting); (4) contract overriding the directive's user exceptions for UNPROTECTED databases. Focus = independently DERIVED numeric streams (shuttle/player coordinates, rally/event timestamps) computed by our own CV/AI from a broadcast video, then redistributed. NOT advice on: broadcast/film copyright in the video signal itself, performers'/sports-organiser commercial rights, unfair-competition/parasitism under national law, or post-Brexit UK law (UK retained a parallel but now-diverging regime). Not legal advice; consult EU IP counsel before relying.",
    "webUsed": true,
    "findings": [
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        "claim": "The sui generis database right protects only investment in OBTAINING (seeking out and collecting pre-existing independent materials), VERIFYING, or PRESENTING contents ? NOT investment in CREATING the data. Resources spent generating new data do not count toward the 'substantial investment' that triggers the right.",
        "support": "In C-203/02 the CJEU held that 'investment in ? the obtaining ? of the contents refers to the resources used to seek out existing independent materials and collect them' and that resources used to CREATE the data (e.g. fixing the list of runners) fall outside this and do not count toward substantial investment under Art 7(1). The parallel Fixtures Marketing line (C-46/02, C-338/02, C-444/02, decided the same day) reached the same conclusion for football fixture lists: investment in generating the data does not qualify.",
        "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill (judgment, 9 Nov 2004)",
        "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203",
        "confidence": "high"
      },
      {
        "claim": "This creating-vs-obtaining distinction is the crux for sports data: a party who is the SOURCE of dynamically-generated sports data (fixtures, race entries, and by extension live match events) often holds NO sui generis right in it, because its spend was on creation, not obtaining. Only investment in collecting/verifying pre-existing materials qualifies.",
        "support": "Commentary on the 2004 quartet (C-46/02, C-203/02, C-338/02, C-444/02) notes the CJEU 'seriously restricted the scope of sui generis protection by admitting that investments in the creation of a database's contents cannot count as investment for obtaining'; the football-fixture and horseracing-runner-list producers therefore FAILED to establish a sui generis right.",
        "sourceName": "Wolters Kluwer / Kluwer Copyright Blog ? analysis of the Database Directive sports-data cases",
        "sourceUrl": "https://legalblogs.wolterskluwer.com/copyright-blog/databases-sui-generis-protection-and-copyright-protection/",
        "confidence": "high"
      },
      {
        "claim": "'Extraction' and 're-utilisation' are construed broadly: any unauthorised appropriation/transfer or making-available of all or part of a protected database's contents, and they do NOT require direct access to the database ? indirect copying (e.g. via a third party who copied from the maker) still counts.",
        "support": "C-203/02 ruling: extraction and re-utilisation 'must be construed as referring to any unauthorised act of appropriation and ? distribution to the public of the whole or a part of the contents of a database', and 'those terms do not imply direct access to the database concerned.'",
        "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill",

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    "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203",
    "confidence": "high"
  },
  {
    "claim": "A 'substantial part' is judged QUANTITATIVELY (volume taken vs. total
contents) OR QUALITATIVELY (scale of the investment in obtaining/verifying/presenting the taken
portion, regardless of its volume). The right protects the investment, not the data as such;
taking data whose creation ? but not obtaining ? was costly does not infringe.",
    "support": "C-203/02 ruling defines quantitative substantiality as 'the volume of data
extracted ? assessed in relation to the total volume of the contents' and qualitative
substantiality as 'the scale of the investment in the obtaining, verification or presentation of
the contents ? regardless of whether that subject represents a quantitatively substantial part.'
Because BHB's investment was in CREATING the data, William Hill's takings were neither
quantitatively nor qualitatively substantial.",
    "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill",
    "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203",
    "confidence": "high"
  },
  {
    "claim": "Art 7(5) anti-circumvention: repeated and systematic extraction/re-utilisation
of INSUBSTANTIAL parts can still infringe, but only where the cumulative effect would
reconstitute and/or make available a substantial part WITHOUT authorisation, thereby seriously
prejudicing the maker's investment. If no protected substantial part exists, systematic small
takings still do not infringe.",
    "support": "C-203/02 ruling: Art 7(5) prohibits acts 'which, because of their repeated
and systematic character, would lead to the reconstitution of the database as a whole or ? a
substantial part ? and/or ? make available to the public ? a substantial part,' i.e. acts that
'seriously prejudice the investment.'",
    "sourceName": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill",
    "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203",
    "confidence": "high"
  },
  {
    "claim": "Database COPYRIGHT (Art 3) protects only the STRUCTURE ? an original SELECTION
or ARRANGEMENT that is the author's own intellectual creation. It never protects the
data/contents themselves, and the intellectual effort/skill spent CREATING the data is
irrelevant to copyright eligibility. Mere significant labour and skill ('sweat of the brow')
cannot ground copyright absent originality in selection/arrangement.",
    "support": "C-604/10 ruling: a database is protected by copyright 'provided that the
selection or arrangement of the data ? amounts to an original expression of the creative freedom
of its author'; 'the intellectual effort and skill of creating that data are not relevant'; and
'significant labour and skill ? cannot as such justify such protection if they do not express
any originality in the selection or arrangement.' The Directive also precludes national
copyright on conditions different from Art 3(1).",
    "sourceName": "EUR-Lex ? Case C-604/10 Football Dataco v Yahoo (judgment, 1 Mar 2012)",
    "sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:62010CJ0604",
    "confidence": "high"
  },
  {
    "claim": "Jurisdiction/localisation: sending data extracted from a protected database
from a server in Member State A to users in Member State B is an act of 're-utilisation'
occurring in B ? but only where there is evidence the operator INTENDED TO TARGET the public in
B (language, currency, local match content, targeting of local customers). Locating a server
outside the EU/a Member State does not by itself escape liability.",
    "support": "C-173/11 ruling (18 Oct 2012): re-utilisation occurs in the Member State of
reception where 'there is evidence ? of an intention on the part of [the sender] to target
members of the public in [that] Member State', considering factors such as data on that
country's matches, contracts with that market's betting firms, and the language used.",
    "sourceName": "IPCuria ? Case C-173/11 Football Dataco v Sportradar (judgment, 18 Oct
2012)",
    "sourceUrl": "https://ipcuria.eu/case?reference=C-173/11",
    "confidence": "high"
  },
  {
    "claim": "Ryanair v PR Aviation: if a database enjoys NEITHER copyright NOR the sui

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generis right, the Directive does not apply to it at all ? so the mandatory user-protection provisions (Arts 6(1), 8, 15, incl. the no-contracting-out rule) do NOT apply, and the owner is free to impose CONTRACTUAL limits (e.g. a screen-scraping/automated-extraction ban in website T&Cs) enforceable under national contract law. So even where IP fails, contract can still bind a party who agreed to the terms.",

"support": "C-30/14 ruling (15 Jan 2015): the Directive 'is not applicable to a database which is not protected either by copyright or by the sui generis right ?, so that Articles 6(1), 8 and 15 of that directive do not preclude the author of such a database from laying down contractual limitations on its use by third parties.'",

"sourceName": "EUR-Lex ? Case C-30/14 Ryanair v PR Aviation (judgment, 15 Jan 2015)",

"sourceUrl": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:62014CJ0030",

"confidence": "high"

}

],

"bottomLine": "For our use case ? numeric streams (shuttle/player coordinates, rally/event timestamps) that we COMPUTE ourselves from a broadcast via CV/AI and then redistribute ? the EU IP exposure on the database-right/copyright axis is LOW, but it is NOT zero, and the real risk lives in contract, not IP. Why low on IP: (1) Sui generis right ? under BHB v William Hill (C-203/02) and the Fixtures Marketing line, the right protects investment in OBTAINING/verifying/presenting pre-existing data, not in CREATING it. Coordinates and event timings we derive frame-by-frame are data we CREATE by our own measurement; we are not seeking out and copying a substantial part of someone else's collected dataset. We extract from the VIDEO (a film/broadcast work), not from the rights-holder's official stats database, so 'extraction of a substantial part' of a protected database is hard to make out, and Art 7(5) systematic-takings liability also requires an underlying protected substantial part. (2) Copyright ? under Football Dataco v Yahoo (C-604/10), copyright protects only an original selection/arrangement (structure), never the raw numbers, and never sweat-of-the-brow; our derived coordinates are facts, not a copied original structure. Where it CAN bite: (a) Contract ? Ryanair v PR Aviation (C-30/14) is the live danger: even with no IP right at all, the source's terms of use / broadcast or feed licence can lawfully ban automated extraction and redistribution and bind us if we accepted them; this is the most likely route to liability and is governed by the licence/T&Cs you click through, not the Directive. (b) If you ever ingest an OFFICIAL data feed/stats database rather than self-deriving from pixels, the obtaining-vs-creating analysis flips toward the rights-holder and substantial-part infringement becomes plausible. (c) Football Dataco v Sportradar (C-173/11) means you cannot dodge EU exposure by server location if you target EU users. Separately, the broadcast VIDEO itself is a protected film/signal ? copying or redistributing the footage (as opposed to derived numbers) is a different, real infringement; our highlight reels live on that axis, not this one. Practical guardrail consistent with the project's licensing discipline: self-derive numeric data from sources you are licensed to process, do not scrape official stats feeds, and read every source's T&Cs/licence for automated-extraction and redistribution clauses before ingesting.",

"caveats": [

"Not legal advice ? these are CJEU holdings synthesized from authoritative sources; get EU IP counsel before relying, especially before any commercial data product.",

"The favourable IP analysis assumes we DERIVE the numeric data ourselves from video (creation), and do NOT copy a substantial part of an official/third-party stats database (obtaining). If the input is a licensed/scraped official feed, the conclusion can reverse.",

"Redistributing the broadcast VIDEO/footage itself (highlight clips) raises separate film/broadcast copyright and related-rights issues NOT analysed here ? this question was scoped to derived numeric streams only.",

"Contract (Ryanair C-30/14) is the dominant residual risk and is fact-specific to each source's terms of use / licence; a click-through or feed licence banning automated extraction or commercial redistribution can bind us even with zero IP right in the data.",

"National-law theories beyond the Directive (e.g. unfair competition / parasitism in some Member States, and protections the Directive expressly leaves to national law per Ryanair) are out of scope and may add exposure.",

"Post-Brexit UK has retained a parallel database-right regime that is beginning to diverge from the EU; UK-specific advice is out of scope here.",

"The C-173/11 operative-ruling detail was verified via IPCuria rather than the primary EUR-Lex text; substance is corroborated across multiple commentaries but the exact wording should be confirmed against the EUR-Lex judgment (CELEX 62011CJ0173) before quoting verbatim."

]

},

{

"question": "q",

```

"scope": "s",
"findings": [
  {
    "claim": "Derived numbers are facts; Indian copyright protects only original
arrangement, not facts, and India has no EU-style sui generis database right, so they are not
copyrightable.",
    "support": "EBC v D.B. Modak 2008 SC; databases protected only as original literary
works.",
    "sourceName": "LSPR; IP Matters",
    "sourceUrl": "https://lawschoolpolicyreview.com",
    "confidence": "high"
  },
  {
    "claim": "DPDP covers only personal data of identifiable individuals; anonymized numbers
are outside it; minors trigger s.9 consent + ban on tracking; s.52 fair dealing is closed, not a
commercial shield.",
    "support": "DPDP s.2(t), s.9.",
    "sourceName": "DPDP Act",
    "sourceUrl": "https://www.dpdpa.com",
    "confidence": "high"
  },
  {
    "claim": "Live risk is the scraping act, not the numbers: ToS + IT Act s.43; OLX v
Padawan allowed private use; ANI v OpenAI undecided. Bottom line: lawful on its face, more
permissive than the EU.",
    "support": "Ikigai Law; ANI v OpenAI reserved.",
    "sourceName": "Ikigai; Lexology",
    "sourceUrl": "https://www.ikigailaw.com",
    "confidence": "medium"
  }
],
"bottomLine": "b",
"caveats": [
  "x"
],
"webUsed": true
},
{
  "question": "When is committing/redistributing derived data a CONTRACTUAL (ToS) or TRADE-
SECRET problem independent of copyright ? and how does owner-recorded vs third-party-sourced
footage change the analysis? Covering platform anti-scraping ToS clauses, dataset
redistribution, the scraping-law landscape (hiQ v. LinkedIn, Van Buren), and the trade-
secret/competitive-moat flip for an owner's own recordings.",
  "scope": "U.S. federal/state law as of mid-2026: ToS-as-contract enforceability (hiQ v.
LinkedIn, Meta v. Bright Data, clickwrap/browsewrap doctrine), CFAA scope (Van Buren), YouTube's
verbatim ToS restrictions as the canonical broadcaster/platform example, dataset-redistribution
risk, and the Defend Trade Secrets Act (18 U.S.C. § 1839) standard for owner-recorded data as a
moat. NOT legal advice; not jurisdiction-specific to India where the indexer owner is based
(Indian Contract Act / IT Act not analyzed here). Copyright is treated only as the contrast
baseline, not analyzed.",
  "findings": [
    {
      "claim": "A platform's Terms of Service is an enforceable CONTRACT, so violating an
anti-scraping clause can be breach of contract EVEN WHERE the CFAA and copyright claims fail ?
this is the central 'contract binds when copyright doesn't' point. In hiQ v. LinkedIn the Ninth
Circuit twice held scraping PUBLIC data likely does not violate the CFAA, yet on remand the
district court (Nov 4, 2022) granted LinkedIn summary judgment on BREACH OF CONTRACT because
hiQ's automated scraping and fake-account 'turker' activity violated LinkedIn's User Agreement;
the case settled with a $500,000 payment and an injunction against future scraping.",
      "support": "LinkedIn's User Agreement barred 'Use manual or automated software, devices,
scripts, robots, other means or processes to access, scrape, crawl or spider the Services.' The
court found hiQ 'breached LinkedIn's user agreement both through its own scraping of LinkedIn's
site and using scraped data, and through turkers' creation of false identities.' Contract law
supplied the enforcement mechanism after the CFAA route closed.",
      "sourceName": "Privacy World (Squire Patton Boggs) ? 'Federal Court Rules in Favor of

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LinkedIn's Breach of Contract Claim'",
  "sourceUrl": "https://www.privacyworld.blog/2022/11/federal-court-rules-in-favor-of-linkedins-breach-of-contract-claim-after-six-years-of-cfaa-data-scraping-litigation/",
  "confidence": "high"
},
{
  "claim": "Van Buren v. United States (2021) narrowed the CFAA: 'exceeds authorized access' covers only accessing files/areas of a system that are entirely OFF-LIMITS to the user, NOT merely using accessible information for a prohibited purpose. This is WHY violating a ToS purpose-restriction (e.g. 'no scraping', 'no dataset-building') is generally NOT a federal computer-crime, pushing platforms toward contract law to police scraping of otherwise-accessible/public data.",
  "support": "6-3 holding: an individual 'exceeds authorized access' only by obtaining information located in areas of the computer 'that are off limits to him.' Van Buren used valid credentials but for a prohibited purpose; that was not a CFAA violation. Policy-based access restrictions (like ToS) thus largely fall outside the CFAA after Van Buren.",
  "sourceName": "Van Buren v. United States, No. 19-783 (U.S. June 3, 2021) ? slip opinion",
  "sourceUrl": "https://www.supremecourt.gov/opinions/20pdf/19-783_k53l.pdf",
  "confidence": "high"
},
{
  "claim": "Whether a ToS BINDS at all turns on the user/account relationship. In Meta v. Bright Data (N.D. Cal., Jan. 23, 2024, Judge Chen) the court granted the scraper summary judgment: Meta's Terms 'do not bar logged-off scraping of public data; perforce [they do] not prohibit the sale of such public data.' A scraper accessing PUBLIC pages while LOGGED OUT is not a 'user' bound by the contract, and once an account is terminated the platform cannot bind the former user to ongoing scraping restrictions. This is the decisive contrast with hiQ, where hiQ HAD accounts/agreed to the User Agreement.",
  "support": "Judge Chen: 'there is a strong and compelling argument that Bright Data was not using Facebook as contemplated by the Terms when it scraped public data while not logged-in.' Meta had deliberately removed older language binding 'non-subscribing visitors,' so only account-holders were bound. The court also refused to read survival clauses as perpetual lifetime restrictions. Meta dismissed its claim rather than appeal.",
  "sourceName": "Eric Goldman, Technology & Marketing Law Blog ? 'Bright Data Scores Major Victory in Web-Scraping Dispute with Meta'",
  "sourceUrl": "https://blog.ericgoldman.org/archives/2024/01/game-on-bright-data-scores-major-victory-in-web-scraping-dispute-with-meta-guest-blog-post.htm",
  "confidence": "high"
},
{
  "claim": "Even when a ToS technically applies, enforceability hinges on contract-formation: courts routinely enforce CLICKWRAP (affirmative 'I agree' click) but are reluctant to enforce BROWSEWRAP (terms merely linked, deemed accepted by continued use) absent reasonable/inquiry notice and manifestation of assent. So a pure logged-out scraper who never clicked 'I agree' may face only a weak browsewrap argument ? whereas an API consumer or account-holder who clicked through is firmly bound.",
  "support": "Courts 'apply traditional principles of contract law and focus on whether the plaintiffs had reasonable notice of and manifested assent.' Clickwrap is 'routinely found enforceable' because of affirmative assent; courts are 'reluctant to enforce browsewrap agreements because users are frequently left unaware that contractual terms were even offered,' enforcing them only with conspicuous notice or 'inquiry notice.'",
  "sourceName": "Crowell & Moring ? 'Recent Court Rulings Provide Warnings on the Use of Browsewrap Agreements'",
  "sourceUrl": "https://www.crowell.com/en/insights/client-alerts/recent-court-rulings-provide-warnings-on-the-use-of-browsewrap-agreements",
  "confidence": "high"
},
{
  "claim": "YouTube's ToS contains explicit, layered anti-extraction restrictions that would bind anyone using the site under account/API terms: it prohibits automated access, harvesting identifying info (faces are named), unauthorized downloading, and commercial reuse ? independent of any copyright analysis. These are the canonical clauses that make scraping/dataset-building a CONTRACT breach for a logged-in or API user even when individual clips might otherwise be fair-use or non-copyrightable.",

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    "support": "Verbatim: '(1) access the Service using any automated means (such as robots, botnets or scrapers) except [robots.txt for search engines or written permission]'; 'collect or harvest any information that might identify a person (for example, usernames or faces)'; 'access, reproduce, download, distribute... or otherwise use any part of the Service or any Content except as expressly authorized by the Service or with prior written permission.'",
    "sourceName": "YouTube Terms of Service ? Permissions and Restrictions",
    "sourceUrl": "https://www.youtube.com/static?template=terms",
    "confidence": "high"
  },
  {
    "claim": "Redistributing a DERIVED dataset escalates ToS/legal exposure well beyond mere collection: internal analytics rarely draws disputes, but reselling/redistributing the database, building a competitor, or training commercial models 'heavily escalates exposure.' So committing a derived dataset to a public repo (a form of redistribution) is materially riskier than transient internal use, and is the act most likely to convert a tolerated scrape into an enforced contract/IP breach.",
    "support": "Commentary distinguishes 'displaying excerpts internally for analysis' from 'redistributing content' and 'training an AI model'; 'building a direct competitor, reselling the raw database, or training commercial AI models heavily escalates exposure.' Note many platform ToS (incl. YouTube) separately bar 'distribute'/redistribution and commercial reuse, so the redistribution step itself can be an independent breach.",
    "sourceName": "PromptCloud ? 'Web Scraping Compliance Challenges' (2026)",
    "sourceUrl": "https://www.promptcloud.com/blog/web-scraping-compliance-challenges-enterprise-guide/",
    "confidence": "medium"
  },
  {
    "claim": "FLIP for OWNER-RECORDED footage: when the corpus is the owner's OWN recordings, the ToS/scraping problem largely disappears (no third party's contract governs your own data) and the analysis shifts to TRADE-SECRET / competitive-moat. Under the Defend Trade Secrets Act, 18 U.S.C. § 1839(3), proprietary recordings/derived data CAN qualify as a trade secret IF (a) the owner takes 'reasonable measures to keep such information secret' AND (b) it 'derives independent economic value... from not being generally known to, and not being readily ascertainable through proper means by' competitors. This is the owner's choice to publish (forfeit the moat) or protect (keep the moat).",
    "support": "Statutory text (18 U.S.C. § 1839(3)) requires both 'reasonable measures to keep such information secret' and that the information 'derives independent economic value, actual or potential, from not being generally known to, and not being readily ascertainable through proper means by, another person who can obtain economic value from the disclosure or use of the information.'",
    "sourceName": "18 U.S.C. § 1839 ? Definitions (Cornell Legal Information Institute)",
    "sourceUrl": "https://www.law.cornell.edu/uscode/text/18/1839",
    "confidence": "high"
  },
  {
    "claim": "The trade-secret moat is fragile and self-destructing on publication: because the DTSA requires both secrecy measures AND value-from-secrecy, committing owner-recorded data (or a derived dataset) to a PUBLIC repo destroys trade-secret status ? once 'generally known' or 'readily ascertainable,' it no longer qualifies. For sports data specifically, courts weigh the value to competitors, the effort/cost to develop it, and the difficulty of others duplicating it; footage of public events is harder to protect than proprietary derived analytics. So for an owner's own corpus the live question is strategic (publish to build community/benchmark vs. keep private to preserve the moat), not a third-party contract breach.",
    "support": "DTSA conditions value on the info 'not being generally known... and not being readily ascertainable.' Sports trade-secret analysis turns on 'the value of the information to the business and its competitors, the amount of effort or money expended in developing the information, and the ease or difficulty with which the information could be properly acquired or duplicated by others' ? and much raw sports footage is publicly observable, complicating protection of anything but genuinely proprietary derived work.",
    "sourceName": "Gowling WLG ? 'Trade secrets in professional sports'",
    "sourceUrl": "https://gowlingwlg.com/en/insights-resources/articles/2023/trade-secrets-in-professional-sports",
    "confidence": "high"
  }
],

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"bottomLine": "Committing derived data is a CONTRACT problem (independent of copyright) whenever the data was obtained under an enforceable Terms of Service that bars scraping/extraction/redistribution/commercial reuse ? and post-Van Buren that contract route is precisely what platforms rely on, because the CFAA no longer reaches mere purpose-violations of accessible/public data. Two thresholds decide whether the ToS actually bites: (1) Was there an account/login or clickwrap 'I agree'? If yes (hiQ; any logged-in or YouTube-API/account user), the ToS binds and scraping + redistributing a derived dataset is breach of contract even if every clip is non-copyrightable or fair use. If the data was scraped purely logged-out from public pages with only a browserwrap link (Meta v. Bright Data), the contract may not bind at all. (2) The redistribution step ? committing the dataset to a public repo ? is the highest-risk act, since most ToS independently bar 'distribute'/commercial reuse and it's what turns tolerated scraping into an enforced breach. CONVERSELY, when the corpus is the OWNER'S OWN recordings, there is no third-party contract to breach and the analysis flips to trade secret: the data is a protectable competitive moat under 18 U.S.C. § 1839 only if kept secret with reasonable measures AND it derives value from not being generally known ? so publishing it (the very act of committing it publicly) forfeits that moat. Net rule for the repo: third-party-sourced derived data that was obtained under a ToS-bound relationship should NOT be committed publicly (contract/redistribution risk that copyright analysis won't catch); owner-recorded derived data is the owner's to publish or protect, where the only thing at stake is the trade-secret moat, not anyone else's contract.",

"caveats": [

"Not legal advice. This is U.S.-centric (Ninth Circuit / N.D. Cal. heavy, federal CFAA/DTSA). The indexer owner is Bangalore-based; Indian law (Indian Contract Act 1872, IT Act 2000, and India's evolving data-protection regime) was NOT analyzed and may treat ToS-formation, browserwrap, and trade secrets differently ? get India-qualified counsel for binding conclusions.",

"hiQ and Bright Data are trial-court / Ninth Circuit outcomes (one settled, one dismissed pre-appeal), not Supreme Court precedent, and partly turned on California contract law and the specific wording/evolution of each platform's terms ? outcomes are fact- and circuit-specific and the law is still unsettled.",

"The 'logged-out public scraping isn't bound by ToS' result is favorable but not universal: it depended on Meta having deleted language binding non-users; a platform whose current ToS expressly binds all visitors, or that puts content behind login/clickwrap (most broadcaster OTT services do), changes the answer. YouTube content via the API or any signed-in/clickthrough flow is squarely contract-bound.",

"Copyright, database rights (e.g., EU sui generis), publicity/privacy rights, and platform technical countermeasures (trespass to chattels, as also found against hiQ) are SEPARATE exposure tracks not fully analyzed here ? a clean contract posture does not by itself clear those.",

"The faces/biometrics restriction in YouTube's ToS and GDPR/India privacy law intersect with this repo's own committed guardrail excluding biometrics/minors; redistributing footage of identifiable people raises privacy exposure beyond the contract/trade-secret axes covered here.",

"One secondary commentary source on redistribution-escalation (PromptCloud) is industry commentary rather than a court or statute, hence medium confidence; the underlying redistribution-bars are anchored in the verbatim YouTube ToS and the hiQ ruling."

],

"webUsed": true

},

{

"question": "When does data DERIVED about people in a video become \"personal data\" (GDPR Art.4 / Recital 26) or special-category biometric data (Art.9), or trigger children's rules (GDPR Art.8 / India DPDP s.9)? Specifically: shuttle coordinates vs person bounding boxes vs pose/gait ? which derived streams are privacy-safe to publish, and how minors change everything.",

"scope": "EU GDPR (Art.4(1), Art.4(14), Recital 26, Art.8, Art.9) with EDPB Guidelines 3/2019 on video devices and ICO/Art.29-WP commentary; India's DPDP Act 2023 (s.2(f), s.9) and DPDP Rules 2025 (Rule 10). Mapped onto the project's three derived streams: inanimate-object (shuttle) coordinates, person bounding boxes, and pose keypoints / gait. This is a legal-risk research memo, not legal advice; no India case law on pose/gait exists yet, and the DPDP Rules were only recently finalized.",

"findings": [

{

"claim": "GDPR only protects information about an identified or identifiable NATURAL PERSON ? data purely about an inanimate object is not personal data on its face. BUT 'relating

to' an object can still make data personal if it singles out / is used about a person (Art.29 WP136 content/purpose/result test). So shuttle (shuttlecock) coordinates are about an object and are not personal data UNLESS they are used to single out or identify a specific player.",

"support": "Art.4(1) GDPR defines personal data as 'any information relating to an identified or identifiable natural person'. WP136 (Opinion 4/2007): data 'relate to' a person via a content OR purpose OR result element, and the CJEU confirms 'data that relate to an object can identify a person.' Shuttle trajectories describe the object's position, not a person's body ? no content/purpose/result link to a specific individual on their own.",

"sourceName": "gdpr-info.eu ? Personal Data; Oxford IDPL / Bird & Bird on WP136 'relates to' test",

"sourceUrl": "https://gdpr-info.eu/issues/personal-data/",

"confidence": "high"

},

{

"claim": "Video footage and ANY derived stream that lets you single out a specific human is personal data, even with no name attached ? the bar for 'anonymous' (escaping GDPR entirely) is very high. Recital 26: identifiability is judged by ALL means 'reasonably likely to be used' (singling out, cost, time, available + foreseeable technology). Pseudonymized data (re-linkable with extra info) is still personal data.",

"support": "Recital 26: principles 'should not apply to anonymous information ... rendered anonymous in such a manner that the data subject is not or no longer identifiable'; identifiability turns on 'all the means reasonably likely to be used, such as singling out.' Pseudonymized data that 'could be attributed to a natural person by the use of additional information' remains personal data.",

"sourceName": "gdpr-info.eu ? Recital 26 (Not Applicable to Anonymous Data)",

"sourceUrl": "https://gdpr-info.eu/recitals/no-26/",

"confidence": "high"

},

{

"claim": "Person BOUNDING BOXES (detect/locate/count people, or categorize age/gender) are ordinary personal data, NOT Art.9 special-category biometric data ? provided the purpose is to distinguish a category of people, not to uniquely identify a named individual. EDPB: 'When the purpose of the processing is ... to distinguish one category of people from another but not to uniquely identify anyone, the processing does not fall under Article 9.'",

"support": "EDPB Guidelines 3/2019 on processing of personal data through video devices, as summarized by Fox Rothschild: video that categorizes without unique identification does not trigger Art.9; Art.9 applies only 'if the controller stores biometric data in order to uniquely identify a person.' Counting/age-gender-estimation examples fall outside Art.9.",

"sourceName": "Fox Rothschild ? Video Surveillance Under GDPR: EDPB Issues Final Guidance (summarizing EDPB Guidelines 3/2019)",

"sourceUrl": "https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance",

"confidence": "high"

},

{

"claim": "Biometric data becomes Art.9 SPECIAL-CATEGORY only when processed 'for the purpose of uniquely identifying a natural person.' Behavioural traits ? gait, typing rhythm, and by extension distinctive movement signatures ? are within Art.4(14) biometric data, and become special-category once used for unique identification. The trigger is PURPOSE + technical processing that enables matching, not the raw modality.",

"support": "Art.9(1) GDPR prohibits processing of 'biometric data for the purpose of uniquely identifying a natural person.' Art.4(14) defines biometric data as data from specific technical processing of 'physical, physiological or behavioural characteristics ... which allow or confirm the unique identification.' ICO/commentary: not all biometric data is special-category ? only if used to uniquely identify; gait is named as a behavioural biometric.",

"sourceName": "gdpr-info.eu ? Art.9 GDPR; ICO biometric-recognition key concepts",

"sourceUrl": "https://gdpr-info.eu/art-9-gdpr/",

"confidence": "high"

},

{

"claim": "POSE KEYPOINTS / GAIT are the genuinely risky derived stream: peer-reviewed work shows gait is a biometric identifier that re-identifies people even with faces obscured, and that pose/skeleton 'anonymization' removes appearance but LEAVES GAIT IDENTITY LARGELY INTACT (80-95% gait-recognition accuracy on anonymized data). So publishing raw skeleton/pose tracks is NOT reliably anonymous, and if used/usable to single out a player it can be personal

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data ? and special-category biometric data if the purpose is unique identification.",
  "support": "Weiß et al., 'Privacy Beyond the Face: Assessing Gait Privacy Through Realistic Anonymization in Industrial Monitoring,' Sensors (2025): 'human pose and gait patterns represent biometric identifiers capable of revealing identity even when the face is obscured'; 'anonymization removes appearance but leaves gait identity largely intact, indicating that pose-driven anonymization is insufficient for privacy protection.'",
  "sourceName": "Weiß et al., Sensors 2025 (PMC) ? Privacy Beyond the Face",
  "sourceUrl": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12788357/",
  "confidence": "high"
},
{
  "claim": "MINORS change the analysis on both axes. EU: Art.8 makes a child's consent to information-society-services processing lawful only at 16+ (Member States may lower to a floor of 13), otherwise parental consent + reasonable verification is required; and any biometric/special-category processing of a child stacks Art.9 on top. India DPDP s.9 is stricter: a 'child' is anyone under 18 (s.2(f)), and BEFORE processing any child's personal data the fiduciary must obtain VERIFIABLE parental/guardian consent, must not cause detrimental effect, and is flatly barred from tracking, behavioural monitoring, or targeted advertising of children.",
  "support": "GDPR Art.8(1): processing of a child's data for ISS is lawful where the child is 16+, else needs parental authorization; Member States may set a lower age 'not below 13'; controller must make 'reasonable efforts to verify.' DPDP s.2(f): child = under 18. DPDP s.9(1) requires 'verifiable consent of the parent ... or the lawful guardian'; s.9(2) bars detrimental processing; s.9(3) bars 'tracking or behavioural monitoring of children or targeted advertising directed at children.'",
  "sourceName": "gdpr-info.eu ? Art.8 GDPR; dpdpa.com ? DPDP Act 2023 Section 9",
  "sourceUrl": "https://gdpr-info.eu/art-8-gdpr/",
  "confidence": "high"
},
{
  "claim": "India's DPDP Rules 2025 (Rule 10) operationalize 'verifiable' parental consent: the fiduciary must use technical/organizational measures to confirm the consent is an identifiable adult (18+) and the parent ? either via identity/age details it already reliably holds, or details voluntarily provided (incl. a virtual token issued by an authorized entity such as a government agency or DigiLocker provider). A bare 'I am over 18' checkbox is not sufficient.",
  "support": "DPDP Rules 2025, Rule 10(1): fiduciary must 'adopt appropriate technical and organisational measures to ensure that verifiable consent of the parent is obtained' and verify via 'reliable details of identity and age ... available with the Data Fiduciary' or details/virtual-token from the individual; 'adult' = completed 18 years.",
  "sourceName": "dpdpa.com ? DPDP Rules 2025, Rule 10",
  "sourceUrl": "https://www.dpdpa.com/dpdparules/rule10.html",
  "confidence": "medium"
},
{
  "claim": "Behavioural biometric identification (which gait/pose enables) is widely treated as Art.9 special-category once used to identify, and regulators are increasingly strict on such 'behavioural' processing ? reinforcing that a movement-signature stream tied to an identified athlete is high-risk to publish without an Art.9 lawful basis (typically explicit consent), and is essentially off-limits for minors under DPDP s.9's behavioural-monitoring ban.",
  "support": "Commentary on Art.9 (GDPR Local; ICO) confirms gait/typing-rhythm are behavioural biometrics that become special-category when processed to uniquely identify; DPDP s.9(3) independently prohibits behavioural monitoring of children regardless of identification purpose.",
  "sourceName": "GDPR Local ? Biometric Data GDPR Compliance; ICO biometric guidance",
  "sourceUrl": "https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/",
  "confidence": "medium"
}
},
"bottomLine": "Triage by what the stream is ABOUT and whether it can single out a person. (1) SHUTTLE coordinates describe an inanimate object ? not personal data, and safe to publish, UNLESS you join them to data that singles out a specific player. (2) PERSON BOUNDING BOXES (detect/locate/count, or estimate age/gender) are ordinary personal data when they single out a human, but are NOT Art.9 biometric data as long as the purpose is to detect/categorize, not

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uniquely identify ? publishing coordinates stripped of any identity link is low-risk; publishing the cropped images is identifying personal data. (3) POSE KEYPOINTS / GAIT are the danger zone: gait is a behavioural biometric that re-identifies people even with faces blurred, and skeleton 'anonymization' provably leaves gait identity intact ? so raw pose/gait tracks are NOT reliably anonymous, are personal data when usable to single out a player, and become Art.9 special-category (needing explicit consent) the moment they are used to uniquely identify. Practical publish/no-publish: shuttle tracks = publish freely; bbox/pose COORDINATES detached from any identity and used only for rally detection = generally OK with normal personal-data hygiene; named player + gait/pose signature, or any face crop = do NOT publish without an explicit-consent Art.9 basis. MINORS flip everything: under India's DPDP (child = under 18) you need VERIFIABLE parental consent (Rule 10 ? real ID/age verification, not a checkbox) before processing ANY child's data, you may not cause detrimental effects, and s.9(3) flatly BANS tracking/behavioural monitoring of children ? which a persistent pose/gait signature arguably is. The cleanest path for footage containing minors is to exclude them from any persisted derived stream beyond inanimate shuttle data, or obtain verified parental consent and keep only de-identified object/coordinate data. This matches the project's existing guardrails: exclude minors from training, treat gait/biometrics as strictly future (GDPR Art.9), and keep a generic schema.",

"caveats": [

"This is a legal-risk research memo from public sources, not legal advice ? counsel sign-off is needed before publishing any human-derived stream commercially, especially anything touching minors.",

"The 'shuttle = not personal data' conclusion holds ONLY while shuttle coordinates are not joined to anything that singles out a player; WP136/CJEU confirm object-data can become personal data via the content/purpose/result link, so a per-player shuttle-track tied to a named athlete could itself be personal data.",

"Whether bounding-box / pose COORDINATES (numbers, no imagery) are 'anonymous' is fact-specific under Recital 26's 'means reasonably likely to be used' test ? in a small, known squad (e.g. a club roster) even coordinates can single out a player, defeating anonymity; the gait evidence shows pose coordinates specifically can re-identify.",

"The EDPB Guidelines 3/2019 text was confirmed via a reputable law-firm summary (Fox Rothschild) because the official PDF could not be parsed by the fetch tool; paragraph-exact wording should be checked against the EDPB original before relying on it.",

"DPDP Rules 2025 detail (Rule 10) is recent and was read from a secondary compilation (dpdpa.com), not the official Gazette ? verify the operative text and any class-of-fiduciary/research exemptions under s.9(4)?(5) before designing the consent flow.",

"No India case law yet interprets pose/gait as biometric/personal data; the India analysis leans on the statutory text of s.9 and the DPDP definition of 'personal data', plus EU-analogous reasoning ? treat the India biometric characterization as reasoned inference, not settled law.",

"Purpose of uniquely identifying' is assessed on intended use: the SAME pose/gait pipeline that is non-special-category for rally detection can flip into Art.9 territory if anyone later repurposes it for player ID ? so the safeguard is purpose-limitation + not persisting identifiable gait signatures, not the modality alone."

],

"webUsed": true

},

{

"question": "For a derived numeric trajectory/feature timeseries: what standards make data \"anonymous\"/non-attributable, and can such a timeseries be re-linked to a specific SOURCE VIDEO by an adversary who re-runs the same detector on candidate public videos and correlates the streams (fingerprinting attack)? How does a PRIVATE/owner-recorded source vs a PUBLIC (broadcast/YouTube) source change achievable non-attributability?",

"scope": "Legal de-identification standards (GDPR Recital 26 \"means reasonably likely\"; CJEU Breyer relative approach; Article 29 WP Opinion 05/2014's singling-out/linkability/inference test; EDPB Guidelines 01/2025; ICO motivated-intruder test; NIST SP 800-188; k-anonymity intuition) plus the re-identification/fingerprinting technical literature (trajectory uniqueness, deterministic linkage/correlation attacks). Scope is the SOURCE-VIDEO linkage threat (linking a timeseries back to which video it came from), NOT identifying the human player. Focus on what non-attributability is actually achievable and under what conditions.",

"findings": [

{

"claim": "The legal test for anonymity is risk-based, not absolute: data is outside data-protection law only when, considering 'all the means reasonably likely to be used' (including by third parties) to single someone out, identification is not reasonably likely ?

judged on objective factors like cost, time and available technology.",

"support": "GDPR Recital 26 states the regulation does not apply to anonymous information, and that to determine identifiability 'account should be taken of all the means reasonably likely to be used, such as singling out, either by the controller or by another person', considering 'all objective factors, such as the costs of and the amount of time required for identification, taking into consideration the available technology at the time of the processing and technological developments.' This is the controlling standard the question names.",

"sourceName": "GDPR Recital 26 (official text, gdpr-info.eu)",

"sourceUrl": "https://gdpr-info.eu/recitals/no-26/",

"confidence": "high"

},

{

"claim": "Identifiability is RELATIVE to the party and the means realistically available to it: the CJEU (Breyer) held data is identifiable where combining it with linking data held by a third party is a means 'likely reasonably to be used', but NOT where that combination is prohibited by law or 'disproportionately difficult in terms of time, cost and man-power.' This is the legal hinge for the private-vs-public distinction.",

"support": "In C-582/14 Breyer the CJEU applied Recital 26 and adopted the relative approach: it is not required that all identifying information be in one party's hands; it suffices if the extra data is held by another person AND combining them is reasonably likely. Crucially the Court said the combination is NOT reasonably likely if 'prohibited by law or ... disproportionately difficult in terms of time, cost and man-power.'",

"sourceName": "Bird & Bird analysis of CJEU C-582/14 Breyer v Germany",

"sourceUrl": "https://www.twobirds.com/en/insights/2016/global/cjeu-decision-on-dynamic-ip-addresses-touches-fundamental-dp-law-questions",

"confidence": "high"

},

{

"claim": "Regulators require anonymisation to defeat THREE distinct risks ? singling out (isolate a record), linkability (connect records about the same subject across datasets), and inference. Re-link-to-source via a re-derived stream is a textbook LINKABILITY attack, so defeating it is part of the anonymity standard, not an extra concern.",

"support": "Article 29 Working Party Opinion 05/2014 (WP216) defines singling out as isolating an individual, linkability as the ability to link at least two records concerning the same data subject (in the same or different databases), and inference as deducing the value of an attribute; it treats robust anonymisation as needing to address all three and warns of residual identification risk.",

"sourceName": "Article 29 Working Party Opinion 05/2014 on Anonymisation Techniques (WP216), European Commission",

"sourceUrl": "https://ec.europa.eu/justice/article-29/documentation/opinion-recommendation/files/2014/wp216_en.pdf",

"confidence": "high"

},

{

"claim": "The EDPB's current position is explicitly context-dependent: the SAME data can be anonymous to a party that lacks the linking information yet personal data to a party that holds or can reasonably obtain it. Non-attributability is therefore a property of the (data, holder, environment) tuple, not of the timeseries alone.",

"support": "EDPB Guidelines 01/2025 on pseudonymisation adopt a context-dependent identifiability approach: data can be personal to one party and anonymised to another depending on linking information each possesses, and one must assess 'any additional information that may enable individual identification, including both information under control of the organisation and information outside of its control.' This is reinforced by the post-SRB reconsideration of anonymisation.",

"sourceName": "Hunton ? EDPB advises on pseudonymisation (Guidelines 01/2025)",

"sourceUrl": "https://www.hunton.com/insights/publications/edpb-advises-on-pseudonymisation-for-gdpr-compliance",

"confidence": "high"

},

{

"claim": "The operational 'reasonably likely' test in practice is the ICO motivated-intruder test: would a reasonably competent, motivated person succeed at re-identification using available data and techniques? It explicitly tells assessors to consider linkability and whether NEW datasets could later be released that increase linkage risk ? i.e. risk is not one-time and

depends on what is correlatable.",

"support": "ICO anonymisation guidance frames effectiveness around whether 'it is reasonably likely that a motivated intruder' could re-identify, weighing linkability, singling out and objective factors (cost, time, available tech); and states risk must be re-assessed when new datasets are released that increase linkability or new recipients gain access.",

"sourceName": "ICO ? How do we ensure anonymisation is effective? (motivated intruder test)",

"sourceUrl": "https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-sharing/anonymisation/how-do-we-ensure-anonymisation-is-effective/",

"confidence": "high"

},

{

"claim": "NIST SP 800-188 frames de-identification as choosing a DATA-SHARING MODEL (publish de-identified data, publish synthetic data, a query interface with de-identification, or a non-public protected enclave) and evaluating disclosure risk before release; no traditional de-identification is risk-free, and formal methods (differential privacy) give stronger guarantees than traditional approaches. This codifies that achievable non-attributability is contingent on the release model.",

"support": "NIST/CSRC summaries of SP 800-188 (Sept 2023) state agencies should 'decide upon a data-sharing model, such as publishing de-identified data, publishing synthetic data based on identified data, providing a query interface that incorporates de-identification, or sharing data in non-public protected enclaves', and should evaluate goals and risks before use; the NIST news page notes the document 'cautions users about the inherent limitations of traditional de-identification approaches when compared to formal privacy methods.'",

"sourceName": "NIST CSRC ? NIST Publishes SP 800-188 (De-Identifying Government Datasets)",

"sourceUrl": "https://csrc.nist.gov/News/2023/nist-publishes-sp-800-188",

"confidence": "high"

},

{

"claim": "k-anonymity gives the right intuition for why a single high-resolution timeseries is hard to anonymise: protection comes from being indistinguishable from at least k-1 other records on the quasi-identifiers. A near-continuous numeric trajectory has so many implicit quasi-identifier dimensions that the equivalence class size collapses toward 1 (a uniqueness/curse-of-dimensionality problem), so a raw per-video stream is effectively a fingerprint.",

"support": "k-anonymity (Samarati & Sweeney) requires each record to be indistinguishable from at least k-1 others on quasi-identifiers; Sweeney showed ~87% of the US population is unique on {ZIP, sex, date of birth}, and high-dimensional data resists k-anonymity. A long numeric trajectory is high-dimensional, so its equivalence class tends to k=1.",

"sourceName": "Sweeney, k-Anonymity: A Model for Protecting Privacy (EPIC mirror)",

"sourceUrl": "https://epic.org/wp-content/uploads/privacy/reidentification/Sweeney_Article.pdf",

"confidence": "high"

},

{

"claim": "Trajectory/mobility timeseries are intrinsically near-unique, so 'just removing the ID' does not anonymise them: four spatio-temporal points uniquely identify 95% of individuals in a 1.5M-person dataset, and uniqueness decays only slowly (~as the 1/10 power) with coarser resolution. A detector's per-frame position/feature series is exactly this kind of high-entropy trace.",

"support": "de Montjoye et al., 'Unique in the Crowd' (Scientific Reports, 2013): in hourly antenna-resolution data for 1.5M people, four spatio-temporal points uniquely identify 95% of individuals, and uniqueness of traces 'decays approximately as the 1/10 power of their resolution' ? coarsening buys little uniqueness reduction. (Full text was paywalled/403/405 on Nature, MIT DSpace and Auckland mirrors; numbers corroborated across multiple secondary summaries of the same Scientific Reports paper.)",

"sourceName": "de Montjoye et al., Unique in the Crowd (Scientific Reports srep01376) ? via Semantic Scholar / Medium summaries",

"sourceUrl": "https://www.nature.com/articles/srep01376",

"confidence": "medium"

},

{

"claim": "The fingerprinting/correlation attack the question describes is real and

feasible against a PUBLIC source: a deterministic detector re-run on the candidate public video reproduces a (nearly) matching stream, and the two series can be aligned by temporal correlation to confirm the link. This is documented as deterministic spatiotemporal entity linking / temporal-signature correlation, and trajectory signatures from ~one week identify >75% of objects.",

"support": "Trajectory-based spatiotemporal entity linking shows ID-stripped trajectories are re-linkable by feature matching, with week-long signatures identifying >75% of individuals; 'Flow Intelligence' extracts temporal motion signatures so blocks across videos are matched by temporal-correlation patterns; such linking attacks are described as deterministic. A re-run of the SAME detector on the same public content is the strongest possible correlation source ? near-identical streams, so a near-deterministic match.",

"sourceName": "Trajectory-Based Spatiotemporal Entity Linking (arXiv 2010.01516); Flow Intelligence (arXiv 2504.11949)",

"sourceUrl": "https://arxiv.org/pdf/2010.01516",

"confidence": "medium"

},

{

"claim": "Whether the timeseries is re-linkable to its source depends entirely on whether a correlatable reference EXISTS and is reasonably obtainable. A PUBLIC source video supplies a permanent, low-cost reference an attacker can re-run the detector against; a PRIVATE/owner-recorded source that is never published has nothing to correlate against, so source-linkage is not 'reasonably likely' for an external adversary ? making strong non-attributability of the SOURCE achievable for the private case but not the public case.",

"support": "Re-identification is fundamentally linkage-driven: privacy risk 'cannot be assessed in isolation' and 'depends heavily on the external data ecosystem and what information might be available for linkage,' and 'de-identification performed today might be broken tomorrow as new auxiliary data sets become available.' Combined with Breyer's 'reasonably likely / not disproportionately difficult' limb, this means: public source = reference exists, linkage reasonably likely; truly private/unpublished source = no reference, linkage by an external adversary not reasonably likely (until/unless the source is later published).",

"sourceName": "OVIC ? The Limitations of De-Identification (data environment / external linkage dependence)",

"sourceUrl": "https://ovic.vic.gov.au/privacy/resources-for-organisations/the-limitations-of-de-identification-protecting-unit-record-level-personal-information/",

"confidence": "high"

}

],

"bottomLine": "Non-attributability is a property of the (data, holder, environment) triple ? not of the timeseries itself. Legally, \"anonymous\" is a risk standard, not an absolute: under GDPR Recital 26, as construed by CJEU Breyer (relative approach) and EDPB Guidelines 01/2025, data is non-attributable only when re-identification is not \"reasonably likely\" given the means realistically available to whoever holds it ? and the Art. 29 WP / ICO frameworks treat linking a record back to its source as exactly the \"linkability\" risk that robust anonymisation must defeat. Technically, a derived per-video trajectory/feature stream is essentially a fingerprint: high-dimensional traces are near-unique (de Montjoye: 4 points -> 95%; k-anonymity collapses to k=1 on rich quasi-identifiers), and a deterministic detector re-run reproduces a correlatable stream that temporal-correlation alignment can match. So the fingerprinting attack the question describes is FEASIBLE ? but ONLY against something to correlate with. That is the whole game on the public-vs-private axis: (1) PUBLIC source (broadcast/YouTube): a permanent, cheap reference exists; an adversary can re-run the same detector on candidate videos and confirm the match. Re-linkage is \"reasonably likely\" and the raw stream should be treated as effectively attributable to its source ? coarsening/feature-reduction reduce but do not reliably eliminate this, and the risk only grows as more public video and better matchers appear. Strong source-non-attributability for raw streams here is NOT achievable; if you need it, you must use a release model that breaks correlatability (heavy aggregation, perturbation/differential privacy, synthetic data, or a non-public enclave per NIST SP 800-188). (2) PRIVATE/owner-recorded source, never published: there is no reference stream in the world to correlate against, so source-linkage by an external adversary is not \"reasonably likely\" ? strong non-attributability of the SOURCE is genuinely achievable for that adversary, and is in fact the default. The caveats: it is conditional and revocable (publish the source later, or leak it, and the link snaps back into \"reasonably likely\"); it protects the source-video link, not necessarily the human (a player still identifiable by other auxiliary data is a separate inference/linkability question); and it is relative to the adversary's auxiliary data ? a party that already holds the source file (or its hash) can always re-link, so \"anonymous to outsiders\" is not \"anonymous to everyone.\"",

```

    "caveats": [
      "Source-linkage (which video did this come from) is a different question from human
      identification (who is the player). A timeseries can be strongly non-attributable to a source
      yet still expose a person via other auxiliary data, or vice versa ? keep the two threat models
      separate.",
      "Private-source non-attributability is CONDITIONAL and not durable: it holds only while
      the source stays unpublished and uncompromised. Publishing the source later, a leak, or an
      adversary obtaining the file/its hash re-enables linkage retroactively (the GDPR/ICO 'reassess
      when new data is released' point).",
      "Identifiability is relative to the adversary: a party that holds the source file, the
      detector + the source, or a stored mapping can always re-link regardless of how 'anonymised' the
      stream looks. 'Anonymous to an external adversary' is not 'anonymous to the controller.'",
      "The de Montjoye 95%/4-points and 1/10-power figures are quoted from the canonical
      Scientific Reports paper but were corroborated only via secondary summaries here ? the primary
      PDF returned 403/405 from Nature, MIT DSpace and the Auckland mirror, so that specific finding
      is marked medium confidence.",
      "NIST SP 800-188's full text did not parse from the PDF; the data-sharing-model taxonomy
      (publish / synthetic / query interface / protected enclave) and 'limitations of traditional vs
      formal methods' framing are drawn from official NIST/CSRC summary pages, which is reliable but
      is a summary, not the verbatim section text.",
      "The relative/'subjective' approach is under active legal flux (post-SRB EDPB
      reconsideration; stakeholders complain 'reasonably likely' is undefined). The direction of
      travel supports the private-vs-public distinction, but the exact threshold for 'reasonably
      likely' is not yet settled in case law/guidance, so treat any bright line cautiously.",
      "Coarsening, quantisation, or low-dimensional feature reduction RAISE the cost of a
      correlation match but do not categorically defeat it against a public source (uniqueness decays
      only ~1/10 power with resolution); only aggregation across many sources,
      perturbation/differential privacy, synthetic data, or withholding the raw stream give a
      defensible non-attributability claim for public-sourced data."
    ],
    "webUsed": true
  }
]

```

SKEPTICAL VERIFICATIONS (JSON):

```

[
  {
    "questionKey": "copyright-facts",
    "centralClaim": "Derived numeric data (per-frame shuttle (x,y) coordinates, rally start/end
    timestamps, audio-onset feature values) does not inherit the source video's copyright: it is
    uncopyrightable facts/measurements under Feist, §102(b), and NBA v. Motorola, and a
    coordinate/timestamp table is not a §101 derivative work (it carries none of the video's
    protected expression); at most a thin compilation copyright attaches to a creative
    selection/arrangement. The act of extraction is a separate question ? intermediate copying can
    infringe (Sega) ? but the fair-use line for non-expressive analytic extraction is \"strongly
    favorable\" (Authors Guild v. Google, Google v. Oracle, 2025 AI-training rulings), so the
    practical rule is: ingest only clean-licensed/owned/permissioned videos and the derived dataset
    is yours to use and publish as uncopyrightable facts.",
    "verdict": "qualified",
    "reasoning": "The narrow doctrinal core is correct and well-supported: facts/measurements
    about a recorded event are uncopyrightable (Feist: \"originality, not sweat of the brow, is the
    touchstone\"; §102(b) excludes any \"discovery\"; NBA v. Motorola: game facts are \"purely
    factual information which any patron of an NBA game could acquire from the arena\"), and a bare
    coordinate/timestamp table reuses none of the video's protected expression, so it is not a §101
    derivative work; compilation protection is \"thin\" (Feist). To that extent the bottom line is
    supported. But the memo is materially OVER-CONFIDENT in ways that could get the client in
    trouble, which is why the verdict is qualified rather than supported.\n\n(1) The fair-use
    defense for the EXTRACTION step is now genuinely contested, not \"strongly favorable.\" The memo
    treats Authors Guild, Google v. Oracle and the 2025 AI rulings as nearly settling the point. But
    Thomson Reuters v. Ross (D. Del. 2025) squarely REJECTED the \"I copied an expressive work only
    to reach unprotected facts/data\" theory: the court held one \"cannot justify copying expressive
    works simply by claiming the goal was to reach unprotected facts,\" distinguished
    Sega/Sony/Oracle as functional-CODE cases (\"those cases are all about copying computer code.
    This case is not.\"), found the use non-transformative, and found Factor 4 harm in a derivative
    market for \"data to train AIs\" even though the plaintiff had not entered that market. A
  }
]

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commercial badminton-analytics product whose dataset competes with a rights-holder's own data/analytics licensing fits Ross's reasoning, not Authors Guild's snippet-index reasoning. The U.S. Copyright Office's May 2025 Part 3 report likewise REJECTS the \"AI training is inherently transformative because non-expressive\" argument and endorses a market-dilution theory of harm under Factor 4. So the extraction-side fair-use outcome is fact-specific and unsettled, with real adverse authority ? overstating it as \"strongly favorable\" invites a client to over-rely on a defense that just failed in the closest 2025 case.\n\n(2) The lawful-source caveat is LOAD-BEARING and unlawful acquisition can independently destroy the defense. Sega's intermediate-copying fair use is conditioned on necessity (\"disassembly... the ONLY means of access to those elements... not protected\") PLUS a legitimate reason ? a real limit, not a blanket analytic pass. Bartz v. Anthropic (N.D. Cal. 2025) holds that even a \"spectacularly transformative\" end use did NOT cure infringement arising from pirated source copies, and that piracy-based liability \"could [not] be undone by later\" lawful use ? a holding that produced an ~\$1.5B settlement. The memo's \"ingest only clean-licensed/owned/permissioned videos\" line is therefore not a footnote; it is the dispositive fact, and the memo's framing risks burying it.\n\n(3) Copyright is NOT the only fence ? the memo's \"facts are free to copy\" framing ignores non-copyright theories that survive precisely BECAUSE the data is uncopyrightable. NBA v. Motorola preserved a narrow five-element hot-news misappropriation tort (costly + time-sensitive info + direct competition + free-riding). hiQ/Van Buren expressly left contract, misappropriation and trespass claims intact even where CFAA fails, and ProCD v. Zeidenberg holds anti-scraping / no-extraction ToS clauses are enforceable and NOT preempted by copyright ? so a \"no data extraction\" license term can ground a breach-of-contract claim even though the numbers themselves are uncopyrightable facts. And the analysis is US-only: a Bangalore-based commercial operation publishing the dataset faces EU/UK sui generis database right exposure (Database Directive Art. 7's \"substantial part\" extraction right), even though BHB v. William Hill narrows it for created (as opposed to obtained) data. None of these non-copyright fences is addressed.\n\nBottom line: the status of the OUTPUT data as uncopyrightable facts is solid; the memo's confidence about the EXTRACTION step's fair-use safety, and its silence on contract/misappropriation/database-right exposure and on the source-acquisition's outcome-determinative role, are the qualifications a skeptical second lawyer must flag.

\"correctedClaim\": \"The derived numeric data itself (shuttle (x,y) coordinates, rally timestamps, audio-onset values) is uncopyrightable FACT/measurement and does not inherit the source video's copyright, and a bare coordinate/timestamp table is not a §101 derivative work ? Feist, §102(b), and NBA v. Motorola support this, with at most a thin compilation copyright in a creative schema. BUT two cautions are essential. First, the legality of the EXTRACTION (transiently copying the video to compute the data) is a separate, currently UNSETTLED question, not a strongly favorable one: Sega makes intermediate copying potentially infringing and limits its fair-use safe harbor to copying that is the only means to reach unprotected elements; Thomson Reuters v. Ross (2025) rejected the \"copy an expressive work just to extract facts/data\" fair-use theory, distinguished the functional-code cases (Oracle/Sega/Sony), and found market harm in a derivative data-for-AI market; and the U.S. Copyright Office (2025) rejected the \"non-expressive therefore inherently transformative\" argument and backs a market-dilution harm theory. Fair use here is fact-specific and competition-sensitive ? if the resulting dataset competes with the rights-holder's own data/analytics market, fair use may fail. Second, the source-acquisition basis is outcome-determinative and unlawful acquisition can independently defeat the defense regardless of how transformative the end use is (Bartz v. Anthropic, 2025; ~\$1.5B settlement). Finally, even where copyright does not reach the facts, NON-copyright fences can: contractual no-extraction/anti-scraping terms are enforceable and not preempted (ProCD), hot-news misappropriation survives in narrow competitive circumstances (NBA v. Motorola), and?relevant to a non-US/commercial operation?EU/UK sui generis database rights may restrict extraction of a substantial part. Practical rule: ingest only clean-licensed/owned/permissioned video AND confirm the license/ToS does not bar data extraction; only then is publishing the derived dataset as uncopyrightable facts reasonably safe ? and even then U.S. fair use for any required copying remains a litigable, not settled, question.\"

\"sources\": [
 {
 \"name\": \"Thomson Reuters Enterprise Centre GmbH v. Ross Intelligence Inc. (D. Del. 2025) ? full opinion (BitLaw)\",
 \"url\": \"https://www.bitlaw.com/source/cases/copyright/Thomson-Reuters-Ross.html\"
 },
 {
 \"name\": \"Perkins Coie ? Fair Use Defense Failed in Thomson Reuters v. Ross (analysis of Factor 1 non-transformative + Factor 4 derivative data-for-AI market)\",
 \"url\": \"https://perkinscoie.com/insights/update/fair-use-defense-failed-thomson-reuters-v-ross-jury-still-out-generative-ai\"
 }
]

```

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    {
      "name": "ArentFox Schiff ? Landmark Ruling on AI Copyright: Fair Use vs. Infringement in Bartz v. Anthropic (pirated copies not cured by transformative use)",
      "url": "https://www.afslaw.com/perspectives/alerts/landmark-ruling-ai-copyright-fair-use-vs-infringement-bartz-v-anthropic"
    },
    {
      "name": "Authors Guild ? Bartz v. Anthropic Settlement (~$1.5B; piracy-based liability)",
      "url": "https://authorsguild.org/advocacy/artificial-intelligence/what-authors-need-to-know-about-the-anthropic-settlement/"
    },
    {
      "name": "Sega Enterprises Ltd. v. Accolade, Inc., 977 F.2d 1510 (9th Cir. 1992) ? intermediate-copying fair use limited to 'only means of access' + legitimate reason (BitLaw)",
      "url": "https://www.bitlaw.com/source/cases/copyright/Sega-Accolade.html"
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    {
      "name": "National Basketball Ass'n v. Motorola, Inc., 105 F.3d 841 (2d Cir. 1997) ? facts uncopyrightable BUT narrow five-element hot-news misappropriation tort survives (Wikipedia summary)",
      "url": "https://en.wikipedia.org/wiki/National_Basketball_Ass%27n_v._Motorola,_Inc."
    },
    {
      "name": "ProCD, Inc. v. Zeidenberg, 86 F.3d 1447 (7th Cir. 1996) ? contractual data-use/anti-extraction restrictions enforceable and not copyright-preempted (Wikipedia)",
      "url": "https://en.wikipedia.org/wiki/ProCD,_Inc._v._Zeidenberg"
    },
    {
      "name": "California Lawyers Association ? hiQ v. LinkedIn: CFAA does not bar scraping, but contract/misappropriation/trespass claims left intact",
      "url": "https://calawyers.org/privacy-law/ninth-circuit-holds-data-scraping-is-legal-in-hiq-v-linkedin/"
    },
    {
      "name": "Mayer Brown ? U.S. Copyright Office (May 2025, Part 3) on AI training and fair use: rejects 'inherently transformative/non-expressive' argument; market-dilution harm",
      "url": "https://www.mayerbrown.com/en/insights/publications/2025/05/united-states-copyright-office-weighs-in-on-fair-use-defense-for-generative-ai-training"
    },
    {
      "name": "EUR-Lex ? Case C-203/02 BHB v. William Hill (sui generis database right; 'substantial part' extraction; investment in obtaining vs. creating)",
      "url": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203"
    },
    {
      "name": "Feist Publications, Inc. v. Rural Telephone Service Co., 499 U.S. 340 (1991) ? facts uncopyrightable; thin compilation protection (Justia)",
      "url": "https://supreme.justia.com/cases/federal/us/499/340/"
    }
  ]
},
{
  "questionKey": "eu-database-right",
  "centralClaim": "For numeric streams (shuttle/player coordinates, rally/event timestamps) that we compute ourselves from a broadcast via CV/AI and then redistribute, the EU IP exposure on the database-right/copyright axis is LOW (because the data is \"created,\" not \"obtained,\" under BHB/Fixtures Marketing, and copyright protects only original structure under Football Dataco v Yahoo), with the real residual risk living in contract (Ryanair v PR Aviation) and on the separate footage/broadcast-copyright axis.",
  "verdict": "qualified",
  "reasoning": "The doctrinal skeleton is accurate and well-sourced: BHB v William Hill (C-203/02) and the Fixtures Marketing quartet do exclude investment in CREATING data; Football Dataco v Yahoo (C-604/10) does limit database copyright to original selection/arrangement and

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excludes sweat-of-the-brow; Ryanair v PR Aviation (C-30/14) does let an owner impose enforceable contractual extraction/redistribution bans even where no IP subsists; and Football Dataco v Sportradar (C-173/11) does defeat the server-location dodge for EU-targeted re-utilisation. On those four pillars the memo is right. But the verdict is "qualified," not "supported," because the memo's SINGLE most load-bearing factual premise is the weakest legal point and is contradicted by the leading on-point authority.

(1) THE "WE CREATE THE DATA" PREMISE IS CONTESTABLE AND LIKELY WRONG ON THE BETTER VIEW. The memo treats frame-by-frame coordinates and event timings as data we "create by our own measurement," dropping us outside the sui generis right. That is precisely the argument the defendant LOST in Football Dataco v Sportradar at the national merits stage. Floyd J held that the "Football Live" compilation of live match events (goals, scorers, penalties, substitutions) attracted the sui generis right as OBTAINED data, reasoning that although the recorded materials did not pre-exist, "there were pre-existing facts, namely the events taking place on the pitch" ? the facts existed when the events occurred, not when recorded ? and, crucially, the recorder did NOT control the events (unlike the fixtures-list maker in BHB, who controlled the data by fixing it). A badminton rally, a shuttle's position, and the moment of a let or a smash are exactly such pre-existing physical facts on the court that we observe and record; we do not control or author them. So a rights-holder can credibly argue our streams are "obtained/recorded," not "created," flipping the BHB analysis toward protectability. The memo's framing that measurement = creation is therefore an overconfident reading a court could reject. This is the qualifier the client most needs to hear.

(2) THE MEMO OMITTS THE CONTROLLING MODERN INFRINGEMENT TEST: CV-Online Latvia (C-762/19, 3 June 2021). This post-dates every case the memo cites and reshapes the substantial-part inquiry: extraction/re-utilisation of a substantial part is prohibited only where it "adversely affects the [maker's] investment ... constitut[ing] a risk to the possibility of redeeming that investment," balanced against the free flow of information and competition. This cuts BOTH ways and the memo cites neither side. It HELPS us (a court must find real harm to the rights-holder's ability to recoup, and our derived numbers from broadcast pixels may not parasitise their official feed). But it also means the analysis is no longer the clean "no protected substantial part = safe" rule the memo implies; it is a fact-intensive, harm-and-competition balancing the memo doesn't engage.

(3) THE MEMO IS EU-ONLY AND MISSES UK POST-BREXIT DIVERGENCE. The UK retained its own sui generis database right (assimilated law) and applies pre-2021 CJEU case law ? including, on its own facts, the Sportradar "obtained match events" reasoning. If the UK is a target market (a likely English-language market for highlight products), exposure there is governed by UK law, which is at least as claimant-friendly on the obtaining/creating point, and "EU exposure is low" understates UK risk. Note too the eligibility asymmetry: a non-UK/non-EEA maker may not even hold the right defensively, but that does nothing to reduce our exposure as a potential infringer of a UK/EEA rights-holder's database.

(4) THE CONTRACT AND FOOTAGE POINTS ARE CORRECT AND, IF ANYTHING, UNDERSTATED. Ryanair is correctly flagged as the live danger. But the memo's "self-derive from sources you are licensed to process" guardrail can be self-defeating: most broadcast/streaming licences and platform T&Cs that would make our INGESTION lawful are exactly the instruments most likely to ban automated extraction and redistribution of derived data ? so the act that cures the footage-copyright problem (a licence) is often the act that creates the contract problem. And the separate film/broadcast-copyright axis (copying/redistributing the footage in highlight reels) is correctly walled off but is the project's largest real exposure; the memo should not let the "IP is low" headline bleed into that axis.

Net: the conclusion "LOW but not zero on IP, real risk in contract and footage" is defensible as a bottom line, but it rests on an overstated creating-vs-obtaining call that the leading authority undermines, omits the controlling 2021 CJEU test, and ignores UK divergence. An over-confident reading ("our numbers are created data, so we're clear on the database right") could get the client sued and lose ? exactly as the Sportradar defendant did.",

"correctedClaim": "For numeric streams (shuttle/player coordinates, rally/event timestamps) we compute from a broadcast via CV/AI and redistribute, EU/UK database-right and copyright exposure is REDUCED but is NOT reliably "low," and should not be characterised as near-zero.

(1) The pivotal premise ? that self-derived coordinates are "created," not "obtained" ? is genuinely contestable: in Football Dataco v Sportradar the live match-event data (goals/scorers) was held to be OBTAINED, because the recorder captured pre-existing facts ("events taking place on the pitch") that it did not control; a badminton rally/shuttle position is the same kind of pre-existing fact we record rather than author, so a rights-holder can argue the sui generis right subsists in our streams. (2) Infringement is now governed by CV-Online Latvia (C-762/19, 2021): even a substantial-part taking is actionable only where it adversely affects the maker's ability to recoup its investment, balanced against information flow/competition ? a fact-intensive test, not a clean exemption. (3) The UK retained its own database right and pre-2021 case law, so UK-targeted products carry at least equal (arguably greater) risk; "EU exposure" understates this. (4) Contract (Ryanair v PR Aviation) and footage/broadcast copyright remain

the dominant exposures ? and note the trap that the licence which legitimises ingestion is often the very instrument that contractually bans extraction/redistribution. Practical guardrail (strengthened): do NOT assume self-derivation defeats the database right; treat self-derived event/coordinate streams as potentially \"obtained\" data; avoid taking from any single rights-holder's match in a way that could reconstitute their dataset or harm their feed's monetisation; read every source's T&Cs for automated-extraction and redistribution clauses; get jurisdiction-specific (EU + UK) advice before redistributing derived numeric streams at scale; and keep the footage-copyright analysis strictly separate and front-of-mind for the highlight reels.",

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"sources": [  
  {  
    "name": "EUR-Lex ? Case C-203/02 British Horseracing Board v William Hill (9 Nov 2004)",  
    "url": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A62002CJ0203"  
  },  
  {  
    "name": "EUR-Lex ? Case C-604/10 Football Dataco v Yahoo (1 Mar 2012)",  
    "url": "https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:62010CJ0604"  
  },  
  {  
    "name": "EUR-Lex ? Case C-30/14 Ryanair v PR Aviation (15 Jan 2015)",  
    "url": "https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:62014CJ0030"  
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  {  
    "name": "ipcuria ? Case C-173/11 Football Dataco v Sportradar (18 Oct 2012)",  
    "url": "https://ipcuria.eu/case?reference=C-173/11"  
  },  
  {  
    "name": "ipcuria ? Case C-762/19 CV-Online Latvia v Melons (3 Jun 2021)",  
    "url": "https://ipcuria.eu/case?reference=C-762%2F19"  
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  {  
    "name": "Bird & Bird ? CV-Online Latvia: CJEU complicates enforcement of database rights  
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    "url": "https://www.twobirds.com/en/insights/2021/uk/cv-online-latvia-cjeu-complicates-  
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  },  
  {  
    "name": "Kluwer Copyright Blog ? CV-Online: access-to-information and competition  
concerns enter the sui generis infringement test",  
    "url": "https://legalblogs.wolterskluwer.com/copyright-blog/access-to-information-and-  
competition-concerns-enter-the-sui-generis-rights-infringement-test-the-cjeu-redefines-the-  
database-right/"  
  },  
  {  
    "name": "Simkins ? Database rights back on the sport radar (Floyd J: live match events  
'obtained' because facts pre-existed on the pitch; recorder did not control events)",  
    "url": "https://www.simkins.com/news/database-rights-back-on-the-sport-radar"  
  },  
  {  
    "name": "Practical Law (Thomson Reuters) ? Court of Appeal: website operator jointly  
liable for infringement of Football Dataco database",  
    "url": "https://uk.practicallaw.thomsonreuters.com/1-524-0061"  
  },  
  {  
    "name": "LexisNexis UK ? Legal protection of databases in the UK (post-Brexit  
assimilated sui generis right; pre-2021 CJEU case law still applied; eligibility asymmetry)",  
    "url": "https://www.lexisnexis.com/en-gb/legal/guidance/legal-protection-of-databases-  
in-the-uk"  
  },  
  {  
    "name": "IViR (Hugenholtz) ? Program Schedules, Event Data and the 'Spin-Off' Doctrine  
under the Database Directive",  
    "url": "https://www.ivir.nl/publicaties/download/spinofffordham.pdf"  
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{
  "questionKey": "india-law",
  "centralClaim": "Derived/scraped numerical data are uncopyrightable facts in India and the activity is \"lawful on its face, more permissive than the EU\": Indian copyright protects only original arrangement (EBC v D.B. Modak), there is no EU-style sui generis database right, DPDP covers only identifiable personal data (so anonymized numbers are out, with minors triggering s.9), and the only live risk is the scraping act under ToS + IT Act s.43 ? with OLG v Padawan allowing private use and ANI v OpenAI undecided.",
  "verdict": "qualified",
  "reasoning": "The core legal taxonomy survives scrutiny: facts as such are uncopyrightable, India has no sui generis database right, and DPDP is gated on identifiability of a natural person. But the confident bottom line (\"lawful on its face, more permissive than the EU\") is materially overstated on three fronts. (1) COPYRIGHT IS NOT AS CLEAN AS CLAIMED. EBC v D.B. Modak (2008 SC) adopted a 'skill and judgment' middle path but did NOT abolish low-threshold protection for COMPILATIONS/DATABASES. Indian High Courts repeatedly protect databases of facts as 'original literary works' under s.2(o) on a modest 'skill, labour and judgment' showing ? Burlington Home Shopping v Rajnish Chibber and V. Govindan v E.M. Gopalakrishna Kone, and critically the OLG order itself, which RULED scraped listings qualified as a protectable 'proprietary database.' So if the 'derived numbers' are extracted from a third party's structured database, copying that selection/arrangement can infringe even though the raw facts are free. The supporting finding also mischaracterizes OLG v Padawan: the case was decided AGAINST the scraper; the 'private use' line is obiter dictum, not a ruling that 'allowed private use.' (2) DPDP IS MORE CONTESTED THAN STATED. Anonymized numbers do fall outside DPDP, but the s.3(c)(ii) 'publicly available data' carve-out is NOT a settled safe harbour for non-anonymized personal data: MeitY (Minister Jitin Prasada) has publicly insisted scrapers 'must still comply with consent and other obligations under the DPDPA,' and the IAPP analysis flags that 'the statute and the government's stance simply do not align.' The s.9 children's regime is also stricter than 'consent + tracking ban' ? s.9(3) is a flat PROHIBITION on behavioural tracking AND targeted advertising of children, with only narrow Draft-Rule-11/Fourth-Schedule exemptions. (3) THE SCRAPING ACT RISK IS LIVE, NOT FACIALLY LAWFUL. The Government has affirmatively argued scraping public data 'may be in breach of section 43' of the IT Act; s.66 layers CRIMINAL exposure (up to 3 years / Rs 5 lakh). Indian courts have not yet ruled on enforceability of anti-scraping ToS, so contractual breach + IT Act liability are genuine untested risks rather than 'lawful on its face.' Finally, ANI v OpenAI is no longer merely 'undecided' ? the Delhi HC RESERVED judgment (approx April 2026); a ruling narrowing fair-dealing/TDM could land at any time. Net: the headline 'not copyrightable / lawful on its face / more permissive than the EU' is defensible only for genuinely anonymized, independently-derived aggregate numbers; it is unsafe advice as stated for scraping a third party's structured dataset, for any non-anonymized personal data, or where minors are involved.",
  "correctedClaim": "Aggregate numbers that are genuinely facts are not themselves copyrightable in India, India has no EU-style sui generis database right, and DPDP applies only to personal data of identifiable individuals ? so independently-derived, anonymized numbers are largely outside copyright and DPDP. BUT the activity is not unqualifiedly 'lawful on its face.' (a) If the numbers are lifted from a third party's structured database/compilation, Indian courts protect such compilations as 'original literary works' on a low 'skill, labour and judgment' threshold (Burlington; Govindan; and the OLG order itself, which held a scraped dataset protectable) ? so copying the selection/arrangement can infringe; the 'private use' remark in OLG v Padawan is obiter and the case was decided AGAINST the scraper. (b) The DPDP 'publicly available data' exemption (s.3(c)(ii)) is contested: MeitY has stated scrapers must still meet DPDPA consent obligations, and s.9(3) flatly prohibits behavioural tracking and targeted advertising of children. (c) The scraping ACT carries live, untested civil and criminal exposure ? breach of anti-scraping ToS plus IT Act s.43 (Government has invoked this) and s.66 (up to 3 years' imprisonment / Rs 5 lakh). (d) ANI v OpenAI judgment is RESERVED (approx April 2026) and could narrow fair-dealing/TDM. Bottom line: lawful only if (i) the numbers are independently derived rather than copying a third party's compilation structure, (ii) any personal data is truly anonymized and no minors are tracked, and (iii) anti-scraping ToS and IT Act exposure are accepted as unresolved risks ? not 'facially lawful.'",
  "sources": [
    {
      "name": "EBC v D.B. Modak (WIPO Lex / Supreme Court of India)",
      "url": "https://www.wipo.int/wipolex/en/judgments/details/2840"
    },
    {
      "name": "Ikigai Law - Legality of data scraping in India (OLG v Padawan; IT Act s.43; ToS; database copyright; private-use dictum)",
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    "url": "https://www.ikigailaw.com/article/263/legality-of-data-scraping-in-india"
  },
  {
    "name": "IAPP - Scraping public data in India: innovation enabler or privacy threat? (Govt s.43 position; DPDP s.3(c)(ii) tension)",
    "url": "https://iapp.org/news/a/scraping-public-data-in-india-innovation-enabler-or-privacy-threat-"
  },
  {
    "name": "CaseMine - Burlington Home Shopping v Rajnish Chibber (database as literary work)",
    "url": "https://www.casemine.com/commentary/in/recognition-of-database-compilation-as-a-literary-work:-analysis-of-burlington-home-shopping-pvt-ltd.-v.-rajnish-chibber-&-anr./view"
  },
  {
    "name": "CaseMine - V. Govindan v E.M. Gopalakrishna Kone (compilation copyright)",
    "url": "https://www.casemine.com/commentary/in/establishing-copyright-infringement-in-dictionary-compilations:-v.-govindan-v.-e.m-gopalakrishna-kone/view"
  },
  {
    "name": "DPDPA s.9 - children's data: verifiable consent + prohibition on behavioural tracking/targeted ads",
    "url": "https://www.dpdpa.com/dpdpa2023/chapter-2/section9.html"
  },
  {
    "name": "CyberPeace - Prohibition of behavioural tracking and targeted advertising for children under DPDP s.9(3)",
    "url": "https://cyberpeace.org/resources/blogs/prohibition-of-behavioral-tracking-and-targeted-advertising-for-children-under-the-dpdp-act-2023"
  },
  {
    "name": "The New Publishing Standard - Delhi HC reserves judgment in ANI v OpenAI (April 2026)",
    "url": "https://thenewpublishingstandard.com/2026/04/01/ani-vs-openai-delhi-high-court-judgment-ai-copyright-india/"
  },
  {
    "name": "S.S. Rana & Co. - Data scraping issues in India (IT Act s.43/s.66 criminal exposure)",
    "url": "https://ssrana.in/articles/data-scraping-issues-india/"
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{

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  "questionKey": "contract-tos-tradesecret",
  "centralClaim": "Committing derived data is a contract problem independent of copyright whenever the data was obtained under an enforceable ToS barring scraping/extraction/redistribution: post-Van Buren platforms rely on contract because the CFAA no longer reaches purpose-violations of accessible/public data. Whether the ToS bites turns on (1) account/login/clickwrap assent (hiQ binds; logged-out browwrap-only per Meta v. Bright Data may not bind) and (2) the redistribution step (committing to a public repo) being the highest-risk act because most ToS independently bar \"distribute\"/commercial reuse. Conversely, owner-recorded footage has no third-party contract to breach, so the analysis flips to trade secret under 18 U.S.C. § 1839 ? secrecy + value-from-secrecy ? and publishing forfeits the moat. Net rule: third-party ToS-sourced derived data should NOT be committed publicly; owner-recorded derived data is the owner's to publish or protect, with only the trade-secret moat at stake.",
  "verdict": "qualified",
  "reasoning": "The core contract machinery is correct and well-sourced ? ToS is an enforceable contract that can bite where CFAA/copyright fail (hiQ's Nov 2022 breach-of-contract summary judgment + $500k consent judgment and injunction, after the CFAA route was foreclosed for public data), the clickwrap/browwrap and logged-in/logged-out distinctions are real (Meta v. Bright Data, Jan. 23, 2024, Judge Chen: Meta's Terms \"do not bar logged-off scraping of public data\"), and redistribution is genuinely the riskiest act. But the bottom line is over-confident in four ways that would get this client in trouble, so it must be QUALIFIED rather than accepted as stated.\n\n(1) VAN BUREN IS OVERREAD. The bottom line asserts \"the CFAA no

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longer reaches mere purpose-violations\" as if the CFAA route is cleanly closed. Van Buren's footnote 8 EXPRESSLY RESERVED \"whether this inquiry turns only on technological (or 'code-based') limitations on access, or instead also looks to limits contained in contracts or policies.\" Commentators (Zwillgen, Cooley/CRS) caution against treating Van Buren as fully closing CFAA enforcement against scrapers who violate ToS; a contract/policy that \"applies to the scope of accessible information itself\" may still ground CFAA liability. So contract is the SAFER bet for platforms, but the federal-crime overlay is not as dead as the bottom line implies ? relevant because CFAA exposure (and its statutory damages/criminal posture) is categorically worse for the client than a breach claim.\n\n(2) THE OWNER-FOOTAGE \"FLIP\" IS THE WEAKEST CLAIM AND IS PARTLY REFUTED. \"When the corpus is the owner's OWN recordings... no third-party contract to breach\" is false for the most likely real-world case: footage SHOT BY THE OWNER AT A BADMINTON EVENT. A ticket is a contract of admission that routinely bars recording/redistribution without written permission, and event-organizer/venue/broadcaster rights (BWF/league terms, Rome Convention rebroadcast rights) attach independently. So owner-shot footage of someone else's event carries live contract AND third-party IP exposure that the \"flip\" waves away. The flip only truly holds for footage the owner shot of THEIR OWN private/non-ticketed sessions.\n\n(3) WRONG GOVERNING LAW FOR THIS CLIENT. The trade-secret analysis is built entirely on US law (18 U.S.C. § 1839 / DTSA). The client is Bangalore-based. India has NO trade-secret statute and NO DTSA ? protection runs through the Indian Contract Act 1872, equity, and the common-law action for breach of confidence (plus IT Act s.72 for electronic records). The two-prong DTSA test is not the operative rule; an Indian court applies breach-of-confidence doctrine. The publication-forfeits-the-moat intuition survives the jurisdiction change, but citing § 1839 as \"the\" rule is a jurisdictional error a skeptical reviewer must flag. Conversely, the platform ToS the client scrapes (YouTube etc.) typically carry US/California choice-of-law and could be litigated there ? so the contract exposure is real even though the trade-secret framing is mis-jurisdictioned.\n\n(4) THE ANALYSIS IGNORES TWO EXPOSURE ROUTES INDEPENDENT OF BOTH CONTRACT AND COPYRIGHT. (a) Personal-data/personality rights: a badminton clip dataset contains identifiable faces. YouTube's own ToS flags \"faces\" as identifying info, and publishing such a dataset implicates India's DPDP Act 2023 (facial data = personal data; consent-based) and personality/publicity rights ? exposure that exists even if every clip is non-copyrightable, fair use, AND no ToS binds. (b) Sui generis database right: in the EU, extraction/reutilization of a substantial part of an investment-heavy database is independently actionable regardless of copyright; a redistributed scraped dataset can infringe even if the underlying items are not works. The bottom line's \"copyright analysis won't catch it ? so it's a contract problem\" framing under-counts these.\n\nNUANCE on damages/remedy (does NOT undercut the bottom line, strengthens it): one might argue a breach claim yields only nominal damages so the contract risk is overstated. That argument FAILS ? the operative remedy in scraping disputes is INJUNCTION plus destruction-of-data orders (hiQ's consent judgment ordered destruction of all source code/data/algorithms derived from scraped profiles), which for a committed public dataset means forced takedown and deletion. So the redistribution-risk point stands and is arguably understated.\n\nCONTRACT-FORMATION refinement: \"any logged-in or account user = bound\" is slightly too categorical even for clickwrap ? courts in 2023-24 have refused to enforce poorly-presented \"sign-in wrap\" terms for lack of conspicuous notice / unambiguous assent. For a YouTube API key or an explicit \"I agree\" account signup the binding is firm; for thinner sign-in flows it is fact-specific.\n\nNet: the practical repo rule (don't publicly commit third-party-ToS-sourced derived data; treat owner-recorded data as the owner's to publish/protect) is DIRECTIONALLY SOUND and the right default. But it must be qualified ? Van Buren did not fully kill the CFAA, the owner-footage flip fails for ticketed/event footage, the trade-secret rule for an India-based client is breach-of-confidence not DTSA, and privacy/personality rights (DPDP) + EU database right are independent exposures the contract-vs-copyright dichotomy misses.

\"correctedClaim\": \"Committing third-party-sourced derived data can be a breach of contract independent of copyright where the data was obtained under an enforceable ToS ? and after Van Buren platforms lean on contract because the CFAA's reach over accessible/public data is narrowed (though Van Buren footnote 8 expressly RESERVED whether contract/policy limits can still ground CFAA liability, so the federal-crime route is narrowed, not eliminated). Whether the ToS binds turns on assent: a logged-in/account/API-key or genuine clickwrap \"I agree\" user is bound (hiQ ? breach-of-contract summary judgment, \$500k consent judgment, injunction and order to destroy scraped-derived data), whereas a purely logged-out scraper of public pages under a browwrap-only link may not be (Meta v. Bright Data). Even \"clickwrap\" is not automatic ? thin \"sign-in wrap\" flows fail for lack of conspicuous notice/assent. Redistribution (committing the dataset to a public repo) is the highest-risk act; the dominant remedy is injunction + forced deletion, not just nominal damages.\n\nThe \"owner's own recordings = no contract, only trade secret\" flip is only partly right. It holds for footage the owner shot of their OWN non-ticketed sessions. It does NOT hold for owner-shot footage of a

third party's badminton event: the admission ticket is a contract that typically bars recording/redistribution, and venue/organizer/broadcaster rights (BWF/league terms; Rome Convention) attach independently of any ToS. And the "trade secret" framing must use the CLIENT'S law: the client is in Bangalore, where there is no DTSA ? protection is via the Indian Contract Act 1872, equity, and common-law breach of confidence (IT Act s.72 for electronic records), not 18 U.S.C. § 1839; publication still forfeits the moat under that doctrine. Critically, the contract-vs-copyright frame misses two exposures that bite regardless of either: (a) personal-data/personality rights ? a face-containing badminton dataset implicates India's DPDP Act 2023 and publicity rights even if no ToS binds and every clip is non-copyrightable; and (b) the EU sui generis database right, which independently bars extraction/reutilization of a substantial part of an investment-heavy database. Net repo rule (sound as a DEFAULT, with the above qualifications): do not publicly commit third-party-ToS-sourced derived data; treat owner-recorded data as the owner's to publish/protect ? but clear ticket/venue/broadcaster terms for event footage, scrub or consent-base identifiable faces under DPDP before any public commit, and analyze "trade secret" under Indian breach-of-confidence law, not the DTSA.",

```
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  {
    "name": "Van Buren v. United States ? analysis confirming footnote 8 reserved whether contract/policy limits ground CFAA liability (Zwillgen)",
    "url": "https://www.zwillgen.com/alternative-data/scotus-rules-cfaa-does-not-contain-purpose-based-restrictions/"
  },
  {
    "name": "Van Buren ? Cooley analysis: Court refused to require code-based barrier; footnote 8 left contract/policy limits open",
    "url": "https://www.cooley.com/news/insight/2021/2021-06-09-us-supreme-court-computer-fraud-abuse-act-van-buren"
  },
  {
    "name": "hiQ v. LinkedIn ? Nov 2022 breach-of-contract summary judgment; Dec 2022 $500k consent judgment, injunction and destruction of scraped-derived data (Privacy World / Squire Patton Boggs)",
    "url": "https://www.privacyworld.blog/2022/12/linkedins-data-scraping-battle-with-hiq-labs-ends-with-proposed-judgment/"
  },
  {
    "name": "Meta v. Bright Data (N.D. Cal. Jan. 23, 2024) ? Terms do not bar logged-off scraping of public data; limited to Meta's specific terms (Quinn Emanuel)",
    "url": "https://www.quinnemanuel.com/the-firm/news-events/client-alert-meta-v-bright-data-significant-decision-for-web-scraping-industry/"
  },
  {
    "name": "Clickwrap vs sign-in wrap enforceability ? 2023-24 courts deny enforcement absent conspicuous notice / unambiguous assent (Hunton, Ninth Circuit)",
    "url": "https://www.hunton.com/privacy-and-information-security-law/its-a-wrap-the-latest-from-the-ninth-circuit-on-sign-in-wrap-agreements"
  },
  {
    "name": "Trade secret law in India ? no statute; protection via Indian Contract Act 1872, equity, common-law breach of confidence, IT Act s.72 (S.S. Rana & Co.)",
    "url": "https://ssrana.in/articles/protecting-trade-secrets-india/"
  },
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    "name": "India DPDP Act 2023 ? facial/biometric data is personal data requiring lawful purpose/consent; implications for publishing identifiable images (K Singhania & Co.)",
    "url": "https://ksandk.com/data-protection-and-data-privacy/regulation-of-biometric-data-under-the-dpdp-act/"
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  {
    "name": "In-venue filming of live sporting events ? ticket as contract of admission, venue/broadcaster rights restricting recording and redistribution (Lexology)",
    "url": "https://www.lexology.com/library/detail.aspx?g=09fe6573-daf6-4fff-bcf0-b9c6ccf41792"
  },
]
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{
  "name": "EU sui generis database right ? extraction/reutilization of substantial part
actionable independent of copyright (europa.eu)",
  "url": "https://europa.eu/youreurope/business/running-business/intellectual-
property/database-protection/index_en.htm"
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{
  "questionKey": "privacy-biometrics",
  "centralClaim": "Triage by what the stream is about and singling-out: shuttle coordinates =
inanimate, not personal data, publish freely (unless joined to identifying data); person
bounding boxes = ordinary personal data when they single out a human but NOT Art.9 unless used
to uniquely identify; pose/gait = danger zone (re-identifies even with faces blurred; raw tracks
not reliably anonymous; Art.9 special-category once used to uniquely identify). Practical:
shuttle = publish freely; detached bbox/pose coordinates for rally detection = OK with normal
hygiene; named player + gait/pose signature or face crop = no publish without explicit-consent
Art.9 basis. Minors flip everything: India DPDP (child = under 18) needs verifiable parental
consent (Rule 10), no detrimental effects, and s.9(3) flatly BANS tracking/behavioural
monitoring of children, which a persistent pose/gait signature arguably is.",
  "verdict": "qualified",
  "reasoning": "The triage framework is directionally sound and matches the project's
guardrails, but a skeptical review surfaces four over-confident statements and one omitted 2025
development. (1) \"Shuttle tracks = publish freely\" is too absolute: the findings' own WP136
content/purpose/result test and Recital 26 \"singling out\" mean that shuttle coordinates
published as part of an identifiable, named match are part of a personal-data processing
operation even though the coordinates alone describe an object ? \"freely\" holds only for
genuinely context-free coordinates. (2) The \"pseudonymized data is always personal data\"
framing relies on the superseded absolute reading of Recital 26; the CJEU appeal judgment EDPS v
SRB (Court of Justice, 4 September 2025, setting aside the General Court's 2023 decision) held
pseudonymised data is NOT necessarily personal for every recipient ? identifiability is
relative/recipient-specific (\"means reasonably likely,\" no \"disproportionate effort\"). This
cuts both ways: it can EXEMPT coordinates a recipient genuinely cannot relink, but the
transferor/controller still treats them as personal and recipient-disclosure/transparency duties
persist from the controller's viewpoint. (3) \"s.9(3) flatly BANS\" overstates the law: DPDP
Rules 2025 Rule 12 + Fourth Schedule create class- and purpose-based exemptions from both s.9(1)
and s.9(3) (healthcare, schools, day-care, child-transport; and real-time location tracking for
child safety). Those exemptions almost certainly do NOT rescue an amateur highlight indexer (so
the ban still bites this client), but \"flatly BANS, no exceptions\" is legally inaccurate.
Moreover \"tracking/behavioural monitoring\" is UNDEFINED in the Act and its breadth is
contested, so \"a persistent pose/gait signature arguably is behavioural monitoring\" is the
defensible CONSERVATIVE reading, not a certainty. (4) The Art.9 \"detect/categorize is fine\"
point (EDPB Guidelines 3/2019) is correct but fragile here: the product's value proposition
trends toward per-player tracking, and the moment pose tracks single out/follow a specific
player across rallies the purpose flips to unique identification and Art.9 attaches ? so \"pose
coordinates detached from identity = generally OK\" is true only under a condition this
architecture is prone to violate. The gait-anonymization research (Weiss et al. 2025) is
correctly cited and strengthens the pose-danger conclusion. Confidence on the DPDP s.9(3)-scope
and gait-as-monitoring conclusions is moderate because the primary statutory text and the
medianama scope article could not be fetched (403); those rest on secondary commentary.",
  "correctedClaim": "Keep the object/person/pose triage, but tighten four points. (a)
Shuttle/object coordinates are not personal data in isolation, but publishing them as part of
identifiable match footage does NOT escape the surrounding personal-data obligations ? \"publish
freely\" applies only to genuinely context-free coordinates, not coordinates tied to a named
match/players. (b) Drop the absolute \"pseudonymised = always personal data\" rule: post EDPS v
SRB (CJEU, 4 Sept 2025) identifiability is relative and recipient-specific, so detached
coordinates a recipient genuinely cannot relink may be non-personal in that recipient's hands ?
but you (the controller) still treat them as personal and owe transparency/recipient-disclosure
duties. (c) Replace \"s.9(3) flatly BANS\" with: s.9(3) prohibits tracking/behavioural
monitoring of children subject to Rule 12 / Fourth Schedule exemptions (which do not cover an
amateur highlight indexer, so the ban does apply to this client); note that
\"tracking/behavioural monitoring\" is undefined, so treating a persistent per-player pose/gait
signature as prohibited monitoring is the conservative, not the certain, reading. (d) Stress
that \"bbox/pose coordinates outside Art.9\" holds ONLY while the system does not single out a
specific player; per-player re-identification across rallies/videos flips the purpose to unique

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identification and triggers Art.9. Practical ladder is otherwise sound: shuttle in isolation low-risk; truly identity-detached bbox/pose coordinates for rally detection OK with normal personal-data hygiene; named player + gait/pose signature or any face crop = do not publish without an explicit-consent Art.9 basis; for footage with minors, exclude them from any persisted derived stream beyond inanimate shuttle data, or obtain Rule-10 verifiable parental consent and keep only de-identified object/coordinate data.",

```
"sources": [
  {
    "name": "Clifford Chance ? Pseudonymized data after EDPS v SRB (CJEU, 4 Sept 2025 appeal judgment)",
    "url": "https://www.cliffordchance.com/insights/resources/blogs/talking-tech/en/articles/2025/09/pseudonymized-data-after-edps-v-srb.html"
  },
  {
    "name": "Osborne Clarke ? CJEU clarifies concepts of personal and pseudonymised data: implications for data sharing",
    "url": "https://www.osborneclarke.com/insights/cjeu-clarifies-concepts-personal-and-pseudonymised-data-implications-data-sharing"
  },
  {
    "name": "DPDP Rules 2025 ? Rule 12 (exemptions from s.9(1)/s.9(3) for Fourth Schedule classes/purposes)",
    "url": "https://www.dpdpa.com/dpdparules/rule12.html"
  },
  {
    "name": "Medianama ? New Data Rules Allow Child Tracking for Safety Without Consent (Rule 12 safety carve-out)",
    "url": "https://www.medianama.com/2025/11/223-dpdp-rules-real-time-child-tracking-without-consent/"
  },
  {
    "name": "EDPB Guidelines 3/2019 on processing of personal data through video devices (Art.9 only on unique identification)",
    "url": "https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf"
  },
  {
    "name": "Weiss et al., Sensors 2025 (PMC) ? Privacy Beyond the Face: gait identity survives pose anonymization",
    "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12788357/"
  },
  {
    "name": "CyberPeace ? Prohibition of Behavioural Tracking and Targeted Advertising for Children under the DPDP Act ('tracking' undefined; s.9(5) exemption power)",
    "url": "https://cyberpeace.org/resources/blogs/prohibition-of-behavioral-tracking-and-targeted-advertising-for-children-under-the-dpdp-act-2023"
  }
],
{
  "questionKey": "deidentification-reid",
  "centralClaim": "Non-attributability is a property of the (data, holder, environment) triple, not of the timeseries itself: under GDPR Recital 26 / Breyer / EDPB 01/2025, \"anonymous\" is a relative risk standard turning on whether re-identification is \"reasonably likely\" given the means realistically available to the holder. A per-video trajectory/feature stream is effectively a fingerprint (de Montjoye: 4 points -> 95%; k-anonymity collapses on rich quasi-identifiers), so a deterministic detector re-run enables a correlation attack ? but only against a correlatable reference. Hence the public/private split: PUBLIC source = a cheap permanent reference exists, re-linkage is reasonably likely, raw streams should be treated as attributable (need a release model that breaks correlatability per NIST SP 800-188); PRIVATE/unpublished source = no external reference, so source-linkage by an external adversary is not reasonably likely and \"strong non-attributability of the SOURCE is genuinely achievable... and is in fact the default,\" subject to caveats (conditional/revocable, protects the source-link not the human, and relative to the adversary's auxiliary data).",
  "verdict": "qualified",
```

"reasoning": "The legal scaffolding and the technical core hold up and are, if anything, REINFORCED by the newest authority the bottom line does not cite: CJEU EDPS v SRB (C-413/23 P, 4 Sept 2025) squarely adopts the contextual/relative approach ? \"pseudonymised data must not be regarded as constituting, in all cases and for every person, personal data\" ? confirming non-attributability is a property of the (data, holder, environment) tuple, exactly as claimed. Recital 26, Breyer's \"reasonably likely / not disproportionately difficult\" limb, WP216's singling-out/linkability/inference triad, ICO's motivated-intruder test, NIST SP 800-188's data-sharing-model framing, Sweeney's k-anonymity, and de Montjoye's 4-points->95% (independently verified via the MIT DSpace copy and multiple secondaries) all check out as cited. So the framework, the fingerprint/linkability characterisation, and the public-source conclusion (reference exists -> re-linkage reasonably likely -> treat as attributable; break correlatability to fix it) are well-supported.\n\nBut three things materially qualify the bottom line and would get a client into trouble if read at face value:\n\n(1) OVERSTATED \"DEFAULT ANONYMOUS\" FOR THE PRIVATE CASE. The single most dangerous sentence is that for a private/unpublished source, \"strong non-attributability of the SOURCE is genuinely achievable... and is in fact the default.\" The same SRB judgment that vindicates the relative approach ALSO holds that for the party holding the source/means ? here the owner who recorded the video and can re-run the same deterministic detector ? the data \"necessarily remained personal data\" and \"remains personal\" for whoever holds or could obtain the means to re-link (paras 76-77, 86). So the stream is NOT \"anonymous by default\"; in the owner's own hands it is personal data, full stop. It is only RELATIVELY non-attributable to a specific external adversary who lacks the source. Calling it \"anonymous\"/\"non-attributable\" as a default ? even hedged ? is precisely the GDPR-classification error the controller's-perspective holding exists to prevent. The bottom line's own caveat (\"a party that already holds the source file can always re-link\") concedes this, but the headline \"default\" framing contradicts the caveat.\n\n(2) THE FORWARD-LOOKING DATA ENVIRONMENT CUTS AGAINST \"NO REFERENCE TODAY = SAFE.\" OVIC's guidance states the auxiliary information available today \"is the absolute minimum that might be available to a determined attacker, now or in the future,\" and the ICO treats anonymisation as not a one-time exercise ? it must be re-assessed as new datasets emerge and techniques advance. The bottom line acknowledges revocability, but its \"in fact the default\" framing inverts the regulator's posture: regulators tell you to ASSUME the environment will grow, not to bank on the current absence of a reference. The private-source link is contingent and the burden is on the holder to keep checking.\n\n(3) NO LIGHT ASSUMPTION THAT THE ADVERSARY LACKS MEANS. SRB (per Skadden, Taylor Wessing) requires the discloser to CAREFULLY DOCUMENT the technical, organisational and legal measures preventing re-identification and to revisit them; \"could obtain\" means count. An over-confident \"private = anonymous\" reading skips that assessment. Note too that SRB was set aside and REMITTED, and it CORRECTED the General Court's recipient-only test by re-centring the controller's perspective for transparency duties ? so the strongest recent case both supports the relative approach and warns against the exact over-confidence at issue.\n\nOne source-confidence note: the specific feasibility figure \">75% of objects from week-long signatures\" (arXiv 2010.01516 / 2504.11949) could not be independently verified ? both PDFs failed to parse ? so that quantitative claim stays low/medium confidence; it is not load-bearing because the de Montjoye uniqueness result and the WP216/OVIC fingerprint-linkability framing already establish near-uniqueness of high-dimensional traces. Also \"deterministic detector re-run\" is somewhat optimistic about the attacker (codec/version/parameter drift makes real-world matches probabilistic, not bit-exact) ? but this strengthens, not weakens, the public-source conclusion (correlation alignment still works), and is irrelevant to the legal core.\n\nNet: the analysis is substantially correct and the public-source conclusion is solid, but the private-source bottom line is over-confident on two legally material points (it is personal data to the holder, never \"anonymous by default\"; and the no-reference-today basis is forward-looking and revocable, not a default). That is a material qualification, not a refutation ? hence \"qualified,\" consistent with the skeptical default given genuine doubt about the \"default anonymous\" phrasing.

"correctedClaim": "Non-attributability is a property of the (data, holder, environment) triple, not of the timeseries itself ? and this relative, risk-based standard is now strengthened by the CJEU's newest ruling, EDPS v SRB (C-413/23 P, 4 Sept 2025): \"personal\" is not an absolute characteristic of data, and the same de-identified stream can be personal to one party yet non-attributable to another, judged by the \"means reasonably likely to be used\" by whoever holds it (Recital 26; Breyer's \"not disproportionately difficult in terms of time, cost and man-power\" limb; EDPB Guidelines 01/2025; ICO motivated-intruder test; WP216's singling-out/linkability/inference triad; NIST SP 800-188's release-model framing). Technically, a per-video trajectory/feature stream is a near-unique fingerprint (de Montjoye: four points uniquely identify 95% of 1.5M people, uniqueness decaying only ~1/10 power with coarsening; k-anonymity collapses toward k=1 on rich quasi-identifiers), so re-running a detector reproduces a correlatable stream that alignment can match ? but only against an existing, reasonably-

obtainable reference.\n\nPUBLIC source (broadcast/YouTube): a permanent, cheap reference exists; an adversary can re-run the detector on candidate videos and confirm the match (correlation, not bit-exact determinism, suffices). Re-linkage is \"reasonably likely,\" so raw streams should be treated as effectively attributable to their source; coarsening/feature-reduction mitigate but do not reliably eliminate this, and risk grows as more public video and better matchers appear. To achieve real source-non-attributability you must use a release model that breaks correlatability ? heavy aggregation, perturbation/differential privacy, synthetic data, or a non-public enclave (NIST SP 800-188). This conclusion is robust.\n\nPRIVATE/owner-recorded source, never published: there is no external reference to correlate against, so source-linkage by an EXTERNAL adversary who lacks the source is not \"reasonably likely.\" But it is wrong to call this \"anonymous by default.\" Three corrections: (a) To the holder/owner ? who recorded the video and can re-run the same detector to re-derive the stream ? the data is and remains PERSONAL DATA; SRB holds it \"necessarily remained personal data\" for the party holding the means, and \"remains personal\" for anyone who has or \"could obtain\" the means to re-link. So the correct claim is RELATIVE non-attributability to a specific external adversary lacking the source, never anonymity in the holder's own hands. (b) It is conditional, revocable, and forward-looking: regulators (ICO, OVIC) treat the data environment as expanding ? today's available auxiliary data is \"the absolute minimum that might be available... now or in the future\" ? so the holder must document the assessment and re-test it over time, not bank on the current absence of a reference; publish or leak the source and the link snaps back to \"reasonably likely.\" (c) It protects only the source-video link, not the human (a player identifiable through other auxiliary data is a separate inference/linkability question), and a party already holding the source file or its hash can always re-link. Practical upshot: do not classify a private per-video stream as \"anonymous\"; treat it as personal data in your own hands that is merely non-attributable to outside parties under documented, periodically-revisited conditions.

```

"sources": [
  {
    "name": "CJEU EDPS v SRB (C-413/23 P, 4 Sept 2025) ? contextual approach analysis
(FPF)",
    "url": "https://fpf.org/blog/rethinking-personal-data-the-cjeus-contextual-turn-in-edps-
vs-srb/"
  },
  {
    "name": "Skadden ? EU Court Clarifies When Pseudonymised Data Is Not Personal Data (EDPS
v SRB): controller retains means -> remains personal; document and revisit",
    "url": "https://www.skadden.com/insights/publications/2025/11/in-a-landmark-decision-eu-
court-clarifies"
  },
  {
    "name": "Taylor Wessing ? Analysis of the CJEU judgment (C-413/23 P): set aside and
remitted; controller-perspective for transparency",
    "url": "https://www.taylorwessing.com/en/insights-and-events/insights/2025/09/analysis-
of-the-cjeu-judgment"
  },
  {
    "name": "GDPR Recital 26 (official text) ? 'all the means reasonably likely to be
used'",
    "url": "https://gdpr-info.eu/recitals/no-26/"
  },
  {
    "name": "Bird & Bird ? CJEU C-582/14 Breyer: relative approach; combination 'prohibited
by law or disproportionately difficult'",
    "url": "https://www.twobirds.com/en/insights/2016/global/cjeu-decision-on-dynamic-ip-
addresses-touches-fundamental-dp-law-questions"
  },
  {
    "name": "Article 29 WP Opinion 05/2014 (WP216) on Anonymisation Techniques ? singling
out / linkability / inference",
    "url": "https://ec.europa.eu/justice/article-29/documentation/opinion-
recommendation/files/2014/wp216_en.pdf"
  },
  {
    "name": "European Law Blog ? EDPB 01/2025 on pseudonymisation: 'pseudonymisation
domain', permeable pseudonymisation/anonymisation boundary",

```

```

    "url": "https://www.europeanlawblog.eu/pub/tfef074h"
  },
  {
    "name": "OVIC ? The Limitations of De-Identification: auxiliary info is 'the absolute
minimum... now or in the future'; digital-fingerprint linkage",
    "url": "https://ovic.vic.gov.au/privacy/resources-for-organisations/the-limitations-of-
de-identification-protecting-unit-record-level-personal-information/"
  },
  {
    "name": "de Montjoye et al., Unique in the Crowd (Scientific Reports 2013) ? 4 points ->
95%; uniqueness ~1/10 power of resolution (MIT DSpace copy)",
    "url": "https://dspace.mit.edu/bitstream/handle/1721.1/88233/de%20Montjoye-2013-
Unique%20in%20the%20crowd.pdf?sequence=1&isAllowed=y"
  },
  {
    "name": "NIST CSRC ? SP 800-188 De-Identifying Government Datasets (release-model
framing; limits of traditional de-identification)",
    "url": "https://csrc.nist.gov/News/2023/nist-publishes-sp-800-188"
  }
]
}
]

```

GROUNDING:

CODEBASE FACTS (verified by the lead before this run):

- eval_baselines/ is a TRACKED (committed) dir of JSON baseline snapshots (regression_baseline.json, heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json, experiment_leaderboard.json). Precedent: committed structured regression files already live in git.
 - backend/eval/regression.py loads a committed baseline, scores DB-stored predictions vs ground-truth intervals, flags P/R drop beyond tolerance, with incommensurability guards (IoU/detector/GT-hash). Predictions come from the DB (needs a pipeline run on the video) -> NOT video-free on the prediction side yet.
 - backend/eval/fusion_golden.py extracts *_golden_features.json from VIDEO+trajectory CSV+labels (needs GPU/WSL). backend/eval/fusion_compare.py & backend/eval/ablation.py RE-SCORE those JSON features on pure CPU with NO video (LOVO deltaF1, bootstrap CI, sign test). This is the EXISTING VIDEO-FREE seam.
 - docs/GOLDEN_DATA_SHARING.md (#175): *_golden_features.json are deliberately kept PRIVATE on OneDrive, OUT of git, because they derive from a 'Proprietary corpus.' Existing skipif litmus tests (tests/test_ablation.py::test_litmus_reproduces_gate_a, tests/test_fusion_compare.py) run ONLY if those features are present in output/ -> currently skipped everywhere.
 - Data contracts: trajectory CSV columns (Frame,Visibility,X,Y); candidate/final rally segments = (start,end) time intervals; multi-signal fusion uses shuttle + person + audio + court-bounds feature columns (weak signals retained, default-OFF). video_id = MD5 of the source file (backend/utils/hashing.py) ? i.e. using it as a public key would FINGERPRINT the source file.
 - Guardrails: commercial-clean licensing MIT/Apache/BSD only; paid LLMs never a training source; EXCLUDE MINORS from the training pool; gait/biometrics strictly FUTURE (GDPR Art 9). Owner is Bangalore/India-based; aims at commercialization and possibly co-authoring a research paper.
 - A SEPARATE workflow is concurrently designing the general video-free fixture mechanics. THIS effort focuses on (a) the LEGAL/IP/privacy status of committing DERIVED structured data PUBLICLY, and (b) a non-attributability + encryption ('secret file') architecture.
- GOAL: let an external contributor on another clone run the rally-segmentation regression suite from COMMITTED structured fixtures (no video present), such that the fixtures (i) cannot be attributed to a specific source video/match/person and (ii) are useless to a public user in isolation.
- KEY CONSTRAINT ALREADY ESTABLISHED BY THE LEAD: a contributor who RUNS the tests necessarily can read the plaintext fixtures + any key on their machine (DRM/analog-hole limit). So secrecy-from-contributor is impossible; the achievable goal is NON-ATTRIBUTABILITY + uselessness-in-isolation, plus encryption only as defense against the NO-KEY public.

2. assistant (2026-06-16T03:01:32.193Z)

I'll synthesize the research findings and skeptical verifications into a coherent memo. Let me first ground myself in the actual codebase facts that are relevant to the legal architecture before writing.

3. user (2026-06-16T03:01:32.928Z)

```
1      # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3      **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4      **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts
(trjectory CSVs, `_golden_features.json`, manifests, baseline snapshots) so the **GPU box**
and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6      ## Why this exists
7      The rally-quality measurement is split across two machines by capability:
8      - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL**
stack ? only the GPU box can run it.
9      - **`backend/eval/fusion_compare.py`** / **`backend/eval/ablation.py`** (the LOVO
**litmus**: ?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the
small `_golden_features.json` ? any session (incl. a CPU dev container) can run them.
10
11     Today those artifacts sit in gitignored `output/` (per-machine) and scattered local
folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs
them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap
with OneDrive.
12
13     ## Canonical shared folder
14     `%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ?
`C:\Users\avidu\OneDrive\rally-annotator\golden-data\`
15
16     OneDrive is present + signed-in on both the GPU box and the dev machine under the same
account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an
overwrite):
17
18     ```
19     golden-data/
20         README.md
21         features/<corpus-tag>/          *_golden_features.json      (small ? the key shared
artifact)
22         trajectories/<corpus-tag>/      *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23         manifests/                     *_lovo_manifest.json
24         baselines/                     leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25     ```
26
27     `` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29     ## The GPU ? CPU workflow
30     1. **GPU box** (videos + CSVs + labels present):
31         ```
32         python -m backend.eval.fusion_golden --manifest <manifest> \
33             --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"
34         ```
35         (or extract to local `output/` then copy the `_golden_features.json` into the shared
folder).
36     2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
37     3. **CPU dev/agent session** points the analysis at the shared folder:
38         ```
39         python -m backend.eval.fusion_compare --features-dir "...golden-
data\features\2026-06-15-n6" --record
40         python -m backend.eval.ablation      --features-dir "...golden-
data\features\2026-06-15-n6" --granularity group
41         ```
42     ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/`
to also light up the `skipif` golden-litmus tests
```



```
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
43
44     ## Optional code ergonomics (in #175, not required)
45     - A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46     - A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
47
48     ## Guardrails
49     - The shared folder lives outside the repo ? never committed. Keep `output/`
gitignored and do not symlink the shared folder into a tracked path.
50     - The features derive from the Proprietary corpus ? keep the OneDrive folder
private (no public share links).
51
52     See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector
setup).
53
```

4. user (2026-06-16T03:01:34.351Z)

```
total 28
drwxr-xr-x 1 avidu 197609    0 Jun 14 16:28 .
drwxr-xr-x 1 avidu 197609    0 Jun 16 01:52 ..
-rw-r--r-- 1 avidu 197609 4696 Jun 14 16:28 experiment_leaderboard.json
-rw-r--r-- 1 avidu 197609 1977 Jun 10 17:43 gemini_wrapper_2026-06-10.json
-rw-r--r-- 1 avidu 197609  689 Jun 10 15:57 heuristic_lovo_n6.json
---
__init__.py
__pycache__
ablation.py
annotations.py
audio
batch.py
calibrate_local.py
calibrate_wasb.py
calibration.py
classifier.py
distill_local.py
eval_partial_labels.py
eval_stats.py
experiment.py
export_wasb_segments.py
fusion_audio.py
fusion_compare.py
fusion_golden.py
gemini_refine.py
gt_loader.py
harness.py
metrics.py
per_user_eval.py
per_user_gate.py
per_user_regression.py
promotion.py
rally_seq_proto.py
regression.py
score_calibration.py
segmentation_metrics.py
serve_contrast.py
served_gate_a.py
splits.py
training.py
windowing.py
```

5. assistant (2026-06-16T03:01:39.777Z)

I have the codebase facts I need. The two key architectural facts are confirmed: ``eval_baselines/`` already commits structured JSON snapshots (precedent), and ``*_golden_features.json`` are deliberately kept private off-git because they derive from a "Proprietary corpus." Let me check one more grounding fact about the feature columns and `video_id` fingerprinting risk.

6. user (2026-06-16T03:01:41.964Z)

""""Pure-CPU LOVO comparison: shuttle-only vs shuttle+person learned fusion on the golden set.

The fusion-P3 measurement. Loads the per-video feature JSONs written by ``fusion_golden.py`` and runs the SHIPPED experiment harness (``experiment.compare``, honest leave-one-video-out) ? no new scoring/labeling math. The two variants differ ONLY by ``feature_names`` (the harness's designed seam): the person features are extra columns the learned LogisticRegression weights, so the held-out ?F1 is an honest answer to "does the person cue help?" ? not a hunch.

Run (no GPU):

```
python -m backend.eval.fusion_compare --features-dir output --record
```

```
from __future__ import annotations
```

```
import argparse
```

```
import glob
```

```
import json
```

```
import os
```

```
from typing import Any, Dict, List
```

```
from backend.eval import experiment
```

```
from backend.eval.eval_stats import paired_delta_ci
```

```
from backend.eval.fusion_golden import PERSON_FEATURE_NAMES, SHUTTLE_FEATURE_NAMES
```

```
def delta_stats(res: Dict[str, Any], metric: str = "f1") -> Dict[str, Any]:
```

```
    """Paired (by video) B?A summary for one metric: mean delta + bootstrap CI + sign test
    (eval_stats, C1 ? "never a bare mean"). Aligns the two variants' per-video held-out scores
    by video name, so the headline ?F1 comes with how-wide + did-it-win-on-enough-videos."""
```

```
    by_a = {p["video"]: p[metric] for p in res["variant_a"]["per_video"]}
```

```
    by_b = {p["video"]: p[metric] for p in res["variant_b"]["per_video"]}
```

```
    vids = [p["video"] for p in res["variant_a"]["per_video"]]
```

```
    return paired_delta_ci([by_a[v] for v in vids], [by_b[v] for v in vids])
```

```
def load_videos(features_dir: str) -> List[experiment.VideoCandidates]:
```

```
    """Load every ``*_golden_features.json`` into a VideoCandidates (intervals + the 10
    feature columns + golden gts). The harness derives labels from gts itself."""
```

```
    videos: List[experiment.VideoCandidates] = []
```

```
    for path in sorted(glob.glob(os.path.join(features_dir, "*_golden_features.json"))):
```

```
        with open(path, encoding="utf-8") as f:
```

```
            d = json.load(f)
```

```
            cands = d["candidates"]
```

```
            intervals = [(float(c["start"]), float(c["end"])) for c in cands]
```

```
            features: Dict[str, List[float]] = {}
```

```
            for fn in SHUTTLE_FEATURE_NAMES:
```

```
                features[fn] = [float(c["shuttle"][fn]) for c in cands]
```

```
            for fn in PERSON_FEATURE_NAMES:
```

```
                features[fn] = [float(c["person"][fn]) for c in cands]
```

```
            gts = [(float(a), float(b)) for a, b in d["gts"]]
```

```
            videos.append(experiment.VideoCandidates(name=d["name"], intervals=intervals,
```

```
=== HASHING ===
```

```
""""Canonical file-identity hashing for the indexer.
```

``video_id`` is the MD5 of the **whole** video file. Historically each endpoint

(``backend/main.py``, ``run_eval.py``, ``scripts/run_phase3_eval.py``,

```
`scripts/run_process_and_stitch.py`)
defined its own `get_file_md5` with a different read-chunk size (64KB / 8KB / 1MB).
The hash values agreed (same bytes hashed), but the duplicated code was a footgun:
any future change to one copy would silently diverge the DB key.
```

```
This module is the single source of truth. Import `compute_video_id` everywhere a
video needs to be keyed.
"""
```

```
from __future__ import annotations
```

```
import hashlib
```

64 KB matches the value used by the live pipeline (backend/main.py) before this was centralized. The chunk size does not affect the resulting digest ? it only bounds memory use while streaming large files.

```
_CHUNK_SIZE = 65536
```

```
def compute_video_id(filepath: str) -> str:
    """Return the MD5 hex digest of the whole file ? the canonical video_id.

    Streams the file in fixed-size chunks so multi-gigabyte recordings never load
    into memory at once.
    """
    hasher = hashlib.md5()
    with open(filepath, "rb") as f:
        for chunk in iter(lambda: f.read(_CHUNK_SIZE), b''):
            hasher.update(chunk)
    return hasher.hexdigest()
```

7. assistant (2026-06-16T03:01:46.501Z)

I have all the grounding I need. The key architectural facts are confirmed:

- The feature JSON carries `name`, `candidates` (start/end intervals + shuttle/person feature columns), and `gts` (ground-truth intervals). The `name` field and any embedded `video\_id` (MD5 of the source file) are the fingerprinting/attribution vectors.
- `eval\_baselines/` is the committed-precedent path; `\*\_golden\_features.json` are the proprietary-corpus artifacts currently kept off-git.

Now I'll synthesize the full memo and return it via StructuredOutput.

8. user (2026-06-16T03:05:36.873Z)

Structured output provided successfully

9. assistant (2026-06-16T03:05:55.297Z)

Memo delivered via StructuredOutput. Summary of what I produced and the grounding behind it:

****The synthesized memo's spine:**** The derived numeric data does not inherit the video's copyright in any of the three jurisdictions (the one settled point) ? but copyright is the easy axis, and the real exposure lives in four other places: the extraction act (now-contested US fair use), contract/ToS, the EU/UK sui generis database right, and privacy. The analysis forks on data provenance (owned/permissioned = Fork A, committable; third-party-sourced = Fork B, not committable), splits by jurisdiction (India primary, US, EU/UK), and gives concrete testable conditions plus a no-publish stream list (person/pose/gait/face/minors).

****Where I integrated the skeptical verifications**** (these were the load-bearing corrections that prevented an over-confident memo):

- US fair use for extraction reframed from "strongly favorable" to contested (Thomson Reuters v.

Ross 2025, USCO May-2025, Bartz v. Anthropic).

- EU "we created it" premise flagged as genuinely contestable (Football Dataco v. Sportradar "obtained" reasoning + CV-Online Latvia harm test + UK divergence).
- India "lawful on its face" downgraded (Indian compilation copyright on a low threshold, OLG decided against the scraper, contested s.3(c)(ii) carve-out, IT Act s.43/s.66).
- Owner-footage "flip" narrowed (ticketed-event footage carries live contract exposure; trade-secret framing corrected from DTSA to Indian breach-of-confidence).
- Privacy "pseudonymised = always personal" and "private = anonymous by default" both corrected via EDPS v. SRB (CJEU 2025).

****Codebase grounding that shaped the engineering conditions**** (relevant file paths, all absolute):

- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN\_DATA\_SHARING.md` ? confirms `\*\_golden\_features.json` are deliberately kept private because they derive from a "Proprietary corpus."
- `C:\Users\avidu\Projects\badminton-highlight-indexer\eval\_baselines\` ? confirms committed structured-JSON precedent already exists in git.
- `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\fusion\_compare.py` ? the feature JSON carries a `name` field plus per-candidate intervals + shuttle/person feature columns + `gts`; the `name`/source-attribution fields are the de-attribution target.
- `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\utils\hashing.py` ? `compute\_video\_id` is the MD5 of the whole source file, which is why I flagged committing `video\_id` as a fingerprinting/source-confirmation leak that must be replaced with an opaque surrogate.

All citations in the memo are drawn only from the findings/verifications provided; none were invented.